

Investigating Properties of Bacterial Genes

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Abbreviations

AIC	Akaike Information Criterion
AUC	Area Under the Curve
bp	Base Pairs
DFRO	Distance From the Replication Origin
DNA	Deoxyribonucleic Acid
ELPD	Expected Log Predictive Density
ESS	Effective Sample Size
FN	False Negatives
FNR	False Negative Rate
FP	False Positives
FPR	False Positive Rate
GC	Guanine Cytosine
HMC	Hamiltonian Monte Carlo
HMM	Hidden Markov Model
INDENS	Insertion Density
LOOCV	Leave One Out Cross Validation
LR	Logistic Regression
MCMC	Markov Chain Monte Carlo
RNA	Ribonucleic Acid
ROC	Receiver Operating Characteristic
TA	Thymine-Adenine
TIS	Transposon Insertion Sequencing
TN	True Negatives
Tn5	Transposon 5
TP	True Positives
TPR	True Positive Rate
TraDIS	Transposon-Directed Insertion-site Sequencing

Lay Summary of the Dissertation

Understanding which genes are essential for bacterial survival is crucial for developing new antibiotics, cancer therapies, and creating useful bacteria for industries such as biofuel. A gene is a sequence of DNA that contains the instructions a cell needs to live, grow, and reproduce. In bacteria, these genes are organized in a circular genome in which a small proportion of the total genes are essential. The remaining genes are non-essential. If a packet of DNA called a transposon is inserted into the genes at different points, the genome will either tolerate this insertion and form a mutant, or die without reproduction. The genomes that do not survive are essential for the organism. Recent advances in genomic research allow the recording of thousands of insertion experiments, creating large libraries of mutant bacteria. This has necessitated the development of mathematical and computational tools to parse this data and classify which genes are essential.

There are two hypotheses about essential genes that, if correct, may improve their classification. The first is that essential genes tend to cluster together within a genome. The second is that essential genes are located closer to the genome's replication starting point, known as the origin. For this report, two bacteria, *Escherichia Coli* (abbreviated to *ecoli*) and *Staphylococcus Aureus* (abbreviated to *saureus*) are analysed for these behaviors. The datasets provide information on which gene is essential or non-essential, its position and length. The total count of transposon insertions into the gene is also provided. The insertion density is calculated as the amount of insertions divided by the size of the gene (as some genes are longer or smaller than others). Essential genes tend to have low insertion densities, as they are intolerant of many transposon insertions. Non-essential genes can tolerate higher insertion densities. This is a common feature in all genomes and may be used to improve the classification of essential genes.

The hypotheses are investigated by measuring two features of essential genes. First, how far each essential gene is from the origin of replication (the mean or median distance) is determined. Second, if the essential genes group together in clusters (the mean or max cluster size). There are different methods to define gene clusters. The "run length" is the number of essential genes which appear consecutively in sequence, forming contiguous chains. These run lengths are found for the real *ecoli* and *saureus* data, after which all the essential and non-essential genes labels are rearranged to form synthetic datasets. These cluster

sizes and distances are calculated for each synthetic dataset. If the synthetic data produces similar results to the real genome data, then there is little evidence to support the hypotheses. This check is performed 10,000 times to see how many results match the real data. The analysis showed there was strong evidence of clustering for all genomes as no simulated data had cluster sizes close to the real data. This was not the same for the distance from the origin, which instead found that the saureus essential genes prefer to be located further away from the origin.

There are different methods to classify essential genes. Some are not suitable due to computational time required, whereas others do not provide much biologically meaningful detail about the results. The classification method selected is called Hidden Markov Models (HMMs), which assumes there are hidden states that we cannot see directly but influence the data that we do observe. The HMM is designed to model a section of the genome consisting of some number of genes. This is done for computational efficiency and to smooth out potential disruptions from bad data. The number of genes in a region that are essential are counted, and modelled using a binomial distribution. It is assumed it is more likely that essential genes are found in essential regions than non essential ones. The probabilities from the regions are taken and averaged over the genes, and this is used to predict the probability of individual genes. The results of the HMM found good results for each of the datasets, with good discrimination between essential and non-essential genes. The model was extended using the insertion density, which was included as an extra piece of information using a Hierarchical Bayes Model. In this structure, each hidden state (essential or non-essential) has global parameters that describe the insertion behavior, but individual regions are allowed to vary randomly around some global values. The results of this model produced similar classification performance but had issues with stable parameter estimation, possibly due to the lack of data. Another model using information on the distance from the replication origin was attempted, but given the weaker signal, this model did not perform well in the classification.

The final part of the report assesses how the models perform on unseen data by simulating datasets that mimic key features of bacterial genomes. Instead of random simulation or splitting into training sets, the datasets were structured to retain the features of the original data such as clustering and lower insertion densities for essential genes. This synthetic genomic data can then be manually adjusted, such as varying the amount of clustering to test how the model responds. The simulations show that the HMM's performance is strongly influenced by spatial clustering with greater clustering leading to better classification, while varying the proportion of essential genes has a smaller effect. Incorporating insertion density improves classification when the distributions of essential and non-essential genes are distinct, but drops with unstable parameters when the signal from the data is weak.

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Chapter 1

Introduction

1.1 Background Biology

The recent emergence of new techniques and methodologies have revolutionized our understanding of bacterial genetics. The consequence of these advances in areas such as high-throughput sequencing and functional genomics has resulted in the generation of vast amounts of genomic data. This has in turn motivated the creation of tools to parse and interpret this data to extract biological insights. Understanding the structure of these bacterial genomes has significant implications in fields such as industrial biotechnology (Giordano, M, et al, 2023), targeted cancer therapeutics (Lian, X, et al, 2025) and the development of antibacterial drugs as inhibitor targets (Hasan, S, et al, 2006).

Bacterial genomes are typically circular and consist of a single, double-stranded DNA molecule known as the chromosome. This chromosome carries the entire genetic blueprint of the organism (genome) which includes all the genes necessary for survival, reproduction, and cellular function. A gene is a specific segment of DNA that encodes functional products and is responsible for conferring traits and regulating biological processes (Baron, S, 1996). In bacterial DNA replication, there is a single defined origin site replicating outward into two halves or arms called replichores (Rocha, E, 2008). Replication eventually ceases at a terminus point (Figure 1.1, adapted from (Shewaramani, 2015)).

1.2 Essential Genes

Genes that support fundamental cellular processes such as DNA replication, transcription, translation, and cellular metabolism by encoding proteins are called “essential” (Fernández-García, G, et al, 2024). They are necessary for the viability of an organism and may be conditionally essential or unconditionally essential. Unconditionally essential genes are those required under all conditions for the reproduction and viability of a cell (Nlebedim, V, et al, 2021). Different organisms may have many conditionally essential genes which require

specific environmental conditions, and these may also confound essential gene identification. Additionally, most genes within a bacterium are non-essential, providing specific enhancements to cell fitness that are beneficial under certain circumstances but not crucial for survival under all conditions (Larrimore, K, et al, 2019).

Since insertion or deletion of essential genes can be lethal to the organism, identifying essential genes is crucial for developing viable therapeutic targets. Studies to compile lists of essential genes have fallen into two general approaches: selective gene deletion to study its function within the genome, and random mutagenesis, where selected mutations are introduced into a genome. Transposon-Directed Insertion-site Sequencing (TraDIS) is a technique which can rapidly build large libraries of mutants through insertions of Tn5 transposons into a genome. A transposon is a DNA segment that can move from one location to another within a genome, which are known as “cut-and-paste” elements or “copy-and-paste” elements (Warner, I, et al, 2023). After transposon insertion, the genes’ function is disrupted which affects its fitness. Essential genes are intolerant of insertions except at extreme positions of the gene (Nlebedim, V, et al, 2021). A Tn5 transposon is a common transposon used for insertion sequencing which inserts randomly in the genome. The selective gene deletion approach is considered a reliable method of assessing gene essentiality, though TraDIS methods are popular due to their high throughput and efficiency when coupled with statistical analysis (Warner, I, et al, 2023).

A number of computational tools have been developed to facilitate the analysis of essential genes from large volumes of TIS data. These include software packages such as ESSENTIALS (Zomer, A, et al, 2012), TRANSIT (Ioerger, T, 2022), ARTIST and EL-ARTIST (Pritchard, J, et al, 2014), each of which integrates genomic information with statistical models to predict gene essentiality. While these tools are effective, there is continued interest in leveraging additional genomic features to improve prediction accuracy. It is hypothesized that essential genes have a tendency to localise close to the point where replication takes place (the origin) and that they exhibit clustering patterns that distinguish them from non-essential genes. Gene expression is the process in which a gene creates a functional product (producing more RNA). It is known that gene expression decreases with increasing distance from the origin, and that the essentiality of a gene can impact its expression (Rocha, E, 2004). Genes that are highly expressed tend closer to the replication origin to avoid collisions with DNA and RNA (Couturier, E and Rocha, E, 2006). Using these features may provide faster and more refined computational methods for the prediction of essential genes.

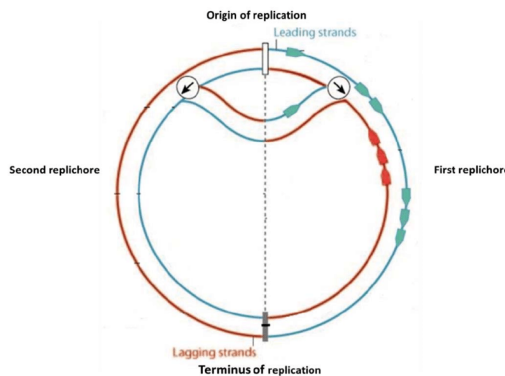


Figure 1.1. Genome organization of *E. coli*.

1.3 Dissertation Objectives

The core aim of this dissertation is to develop approaches which assess the hypothesis that essential genes tend to cluster together and that they exhibit a preference for residing near the replication origin. These techniques are then to be implemented on simulated data and two available bacterial datasets, *E. coli* and *S. aureus*. A third dataset, *S. Typhimurium*, was not available for assessment. The objectives of this dissertation are:

- To perform exploratory data analysis of the two bacterial datasets for the two main research questions for all replichores.
- Propose a method to assess whether essential genes have a tendency to reside near the replication origin and simulate some basic data to investigate the effectiveness of the proposed method. Apply the method to the two datasets and report the results.
- Propose a method to assess the evidence for essential gene clustering. Simulate data to investigate the effectiveness of the proposed method. Apply the method to the two datasets and report the results.
- Consider how to use essential gene clustering and the tendency of essential genes to reside near the replication origin alongside the gene-level insertion densities to classify bacterial genes. Implement the proposed approaches on some of the datasets, depending on the outcomes of the exploratory analysis.

1.4 Data Analysis Software

This dissertation was developed in the R programming language (R Core Team, 2021). The data analysis and modelling was conducted using a range of different packages: ‘tidyverse’ (Wickham, H, et al, 2019), ‘kableExtra’ (Zhu, H, 2024), ‘circlize’ (Gu, Z, et al, 2014), ‘knitr’ (Xie, Y, 2025), ‘scales’ (Wickham, H, et

al, 2025), ‘ggrepel’ (Slowikowski, K, 2024), ‘patchwork’ (Pedersen, T, 2025) and ‘cowplot’ (Wilke, C, 2025). The ‘rstan’ (Stan Development Team, 2025) package was used to develop the Bayesian HMM models, the ‘loo’ (Vehtari, A, et al, 2024) package was used to cross validate the models and ‘caret’ (Kuhn, M, 2008) package for the classification analysis. Other packages used in the modelling analysis include ‘bayesplot’ (Gabry, J and Mahr, T, 2025) and ‘pROC’ (Robin, X, et al, 2011).

1.5 Structure of Dissertation

This dissertation consists of 6 chapters, including the Introduction and Appendices. The structure of the chapters are:

Chapter 2 provides detail on the two datasets used in the dissertation. It follows with exploratory data analysis with regard to the two main research questions of whether essential genes cluster and whether they reside closer to the replication origin. There is also some preliminary data manipulation and preprocessing both for the data analysis and subsequent modelling and testing approaches.

Chapter 3 provides an overview of the analytical techniques used in this dissertation, beginning with Monte Carlo methods for data simulation and their role in permutation-based hypothesis testing. It then introduces Hidden Markov Models (HMMs), outlining their structure and relevance to sequential data analysis, supported by examples from the literature. There is also a brief discussion of classification metrics, relevant for the modelling completed in Chapter 4. Finally, the chapter briefly describes Bayesian theory and its connection to both Markov Chain Monte Carlo methods and HMMs.

Chapter 4 applies the methods from the previous chapter to the *ecoli* and *saureus* datasets, presenting key assumptions, interpreting outcomes, and exploring refinements to the classification. Model performance is evaluated through various validation techniques.

Chapter 5 summarizes the dissertation’s methods and findings, discusses challenges encountered during analysis, and outlines directions for future work.

Chapter 6 provides references and non-critical details left out of the main body of the dissertation.

Chapter 2

Exploratory Data Analysis

2.1 Data Description

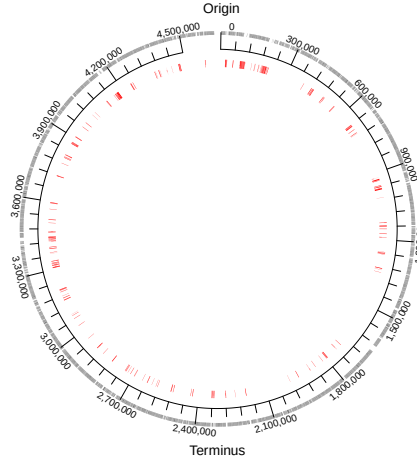
The two datasets, *Escherichia Coli* (abbreviated to *ecoli*) and *Staphylococcus Aureus* (abbreviated to *saureus*), are both derived from the European Nucleotide Archive under accession number ERR2249109 and accession numbers SRR105406 to SRR1056422. It consists of a list of gene insertions using TraDIS for *Ecoli* K-12 strain BW25113, with genes that are considered essential those that survived selective gene deletions and non-essential genes those that had a surviving genome mutant after gene deletion (Baba, T, et al, 2006). For the *saureus* dataset, essential genes were identified from similar genes in different organisms or strains that were reported as essential. If the genes survived with mutations, they were classed as non-essential (Fey, P, et al, 2013) (Chaudhuri, R, et al, 2009). The *ecoli* dataset contains 364,813 observations, with the variable definitions summarized in Table 2.1. The *saureus* dataset shares the same variable names but comprises 140,114 observations. The final index numbers for the *ecoli* and *saureus* datasets are 448,854 and 280,250, respectively, indicating that some rows are missing within each dataset. No null or incorrectly labeled entries were detected. There are 1093 duplicated rows for the *ecoli* dataset, and 426 duplicated rows for the *saureus* dataset.

2.2 Data Preprocessing

Each dataset has multiple transposon insertions per gene. We retain a single distinct observation from each unique gene insertion to minimize the statistical influence of repeated insertions on the same gene. We use gene-level insertion densities, defined as the number of unique transposon insertions normalized by gene length, for modeling in Chapter 4. There is an assumption that transposons insert randomly within the genome, and that the number of unique insertions for a given gene increase correspondingly with the length of the gene. There is therefore a need to scale the unique insertions by the length of the gene

Table 2.1. Column Variable Description and Their Data Type

Column	Description	Data Type
Gene	The identifier for the gene, e.g., 'BW25113-0001', which represents a specific gene in the bacterial strain BW25113.	Character
Position	The position within the gene or genome, referring to the exact nucleotide or base pair location.	Integer
Length	The total length of the gene in base pairs.	Integer
Start	The starting nucleotide position of the gene on the genome.	Integer
End	The ending nucleotide position of the gene on the genome.	Integer
Status	Indicates whether the gene is essential or non-essential.	Character
Count	The number of transposon insertions found in the gene.	Integer

**Figure 2.1.** Circular plot of the start positions and end positions of the Ecoli dataset. Essential genes are shown in red in the inner ring while non-essential genes are shown in grey in the outer. Non-essential genes greatly outnumber essential ones, as only a subset of genes are crucial for biological function.

(Nlebedim, V, et al, 2021). Let g denote a gene and let:

$$D_g = \frac{N_g}{L_g} \quad (2.1)$$

where: D_g = gene-level insertion density, N_g = number of unique transposon insertions in gene g , L_g = length of gene (in base pairs (bp)). Insertion density may then be used as a classifier of essential genes.

Due to the nature of the TraDIS technique, insertions are counted in sequential genomic order rather than being indexed by replicore. The data follows a con-

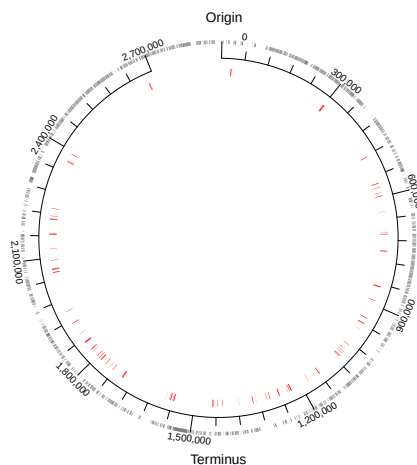


Figure 2.2. Circular plot of the start positions and end positions of the Saureus dataset. Essential genes are shown in red in the inner ring while non-essential genes are shown in grey in the outer. Non-essential genes outnumber essential ones like Ecoli though there are fewer genes overall in this dataset.

tinuous, circular representation counting insertions linearly around the entire genome. It does not segment the counts by replichore by measuring outward from the origin to the terminus in both directions, which would reflect the biological reality. In order to properly reflect this replication behaviour in the data, it is split at the midpoint, which is defined as half the maximum gene end position of the data rounded up to the nearest gene. The left replichore contains genes above the midpoint, and the right replichore contains genes equal to or less than the midpoint. For both datasets, gene positions on the left replichore are rescaled to reflect their relative orientation. This is achieved by reversing the genomic coordinates: the new start position is calculated by subtracting the original gene end position from the total genome length, and the new end position is calculated by subtracting the original gene start position from the genome length. This transformation flips the replichore so that the last gene becomes the first. The absolute value is taken for any negative values. Mathematically, let L be the total genome length, and let each gene have an original start position s and end position e . The transformed coordinates for genes on the left replichore are defined as:

$$s_{new} = |L - e|$$

$$e_{new} = |L - s|$$

where s_{new} is the new start position, e_{new} is the new end position, with constraints $s, e \in \mathbb{N}$ and $0 < s < e < L$.

There is a slight imbalance in the *ecoli* data after processing. There is one additional gene in the left replichore compared to the right. The *saureus* dataset has an equal number of observations on each replichore. Across both datasets, non-essential genes significantly outnumber essential ones: 94% in *Ecoli* and 93% in *Saureus*. Although *Ecoli* has more essential genes (245 vs. 90), its dataset is also much larger (Table 2.2). Split by replichore, the counts and proportion of essential genes are similar except for *ecoli* essential genes, where there are more in the left replichore than the right (139 vs. 106). To preserve the genomic structure of each replichore and avoid distortions in the analysis, each replichore of each bacterium (*ecoli* and *saureus*) is treated as separate datasets. Pooling the replichore data would obscure the positional relationships. For example, if essential gene positions are analyzed together, clusters may appear that span both replichores despite the genes being biologically distinct and located on opposite sides of the origin. Even if a replichore variable (e.g. “left” or “right”) is introduced to adjust for this, the analysis would effectively treat the replichores separately, reinforcing the decision to keep them distinct.

Table 2.2. Gene counts for the full genome as well as per replichore. The table also shows the proportions of genes for the entire genome of each bacterium. There are similar proportions of essential genes across bacterium, except for the left replichore of the *ecoli* dataset, which has more essential genes.

Dataset	Essentiality	Replichore (Count)		Full Genome	
		Left	Right	Count	Proportion
<i>Ecoli</i>	Essential	139	106	245	0.06
<i>Ecoli</i>	Non-Essential	1855	1877	3742	0.94
<i>Saureus</i>	Essential	44	46	90	0.07
<i>Saureus</i>	Non-Essential	636	632	1268	0.93

2.3 Distance from the Replication Origin

In this section, an exploratory analysis is performed on the spatial distribution of essential genes relative to their distance from the replication origin. The replication process influences the organisation of the genome and it is hypothesized that the essential genes will reside closer to the origin. Figure 2.3 shows violin plots overlaid with boxplots of the datasets, split by replichore and gene status. Starting with the *ecoli* dataset, we note that there is a strong preference within the right replichore towards the origin. This is shown with the bulge around 250,000 bp. This tapers afterwards, and in contrast to the non-essential genes, shows a large drop off further away from the replication origin. For the left replichore, there is a weaker preference to the origin with a peak close to half the distance away from the origin. Comparing medians reported on the boxplots,

the left replichore shows roughly similar distances whereas the right replichore median distance is less than half the essential genes. This suggests that essential genes in *ecoli* are located closer to the replication origin, but this behavior is not the same for both replichores.

In the *saureus* dataset, essential genes tend to be located further from the replication origin on both replichores. This is reflected in the violin plots, where the distributions of essential genes are skewed past the midpoint, and their medians are higher than those of the non-essential genes (Figure 2.3). For the non-essential genes, the left replichore exhibits a tendency to cluster nearer the origin, highlighted by a pronounced peak at the origin. The *saureus* plots provide evidence to contradict the original hypothesis that the essential genes are generally situated nearer the origin. This discrepancy shows the distinct positional behavior of essential genes in *ecoli* and *saureus* genomes, highlighting bacterium differences in genome organization.

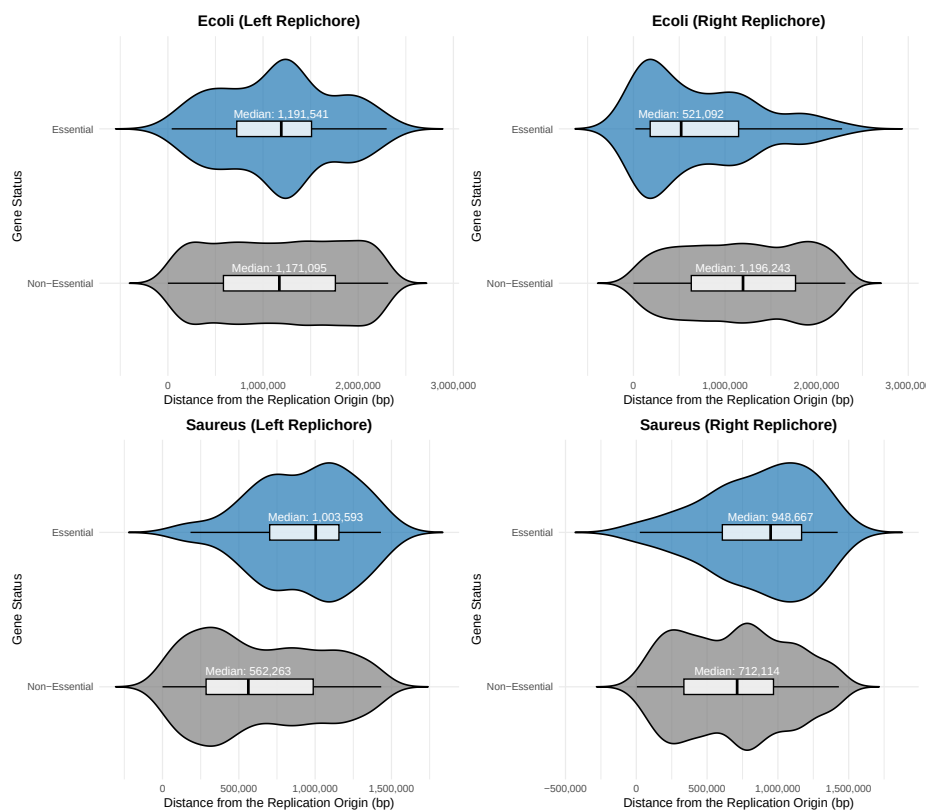


Figure 2.3. Violin plots of each bacterium, split by replichore and essential status. Box plots are overlaid with the median gene distance. There is a bulge for the right replichore of *ecoli*, which is not seen for the left replichore. The *saureus* dataset shows positional preference away from the origin, contradicting the hypothesis.

To further investigate the preference for residing near the replication origin, empirical cumulative distribution function plots are shown in Figure 2.4. This

shows the cumulative proportion of genes (either essential or non-essential) for each of the replichores of the bacterium. We note that for the left replichore of *ecoli*, roughly 50% (0.5) of the essential genes are located before the median distance from the replication origin which indicates no strong preference for the origin. For the right replichore more than 75% (0.75) of the essential genes are located before the median distance. There are strong increases in cumulative proportion ($\sim 40\%$) before the lower quartile of the replication distance, confirming a strong preference for residing near the origin. The cumulative proportion of non essential genes is almost linear for the non-essential genes, highlighting no preference for the replication origin. In the *saureus* dataset, the cumulative proportion of essential genes only reaches 50% in both cases after the distance is $\sim 75\%$ (0.75) from the origin. Further, the cumulative proportion of essential genes just reaches over 30% (0.3) at almost half the distance away from the origin. The conclusion from these plots shows that the hypothesis of preferential position near the origin is only supported for the right replichore of the *ecoli* genome. Additionally, the left replichore of *ecoli* provides only weak or no evidence of preference, while both replichores of the *saureus* dataset exhibit the opposite trend and suggest a preference away from the origin.

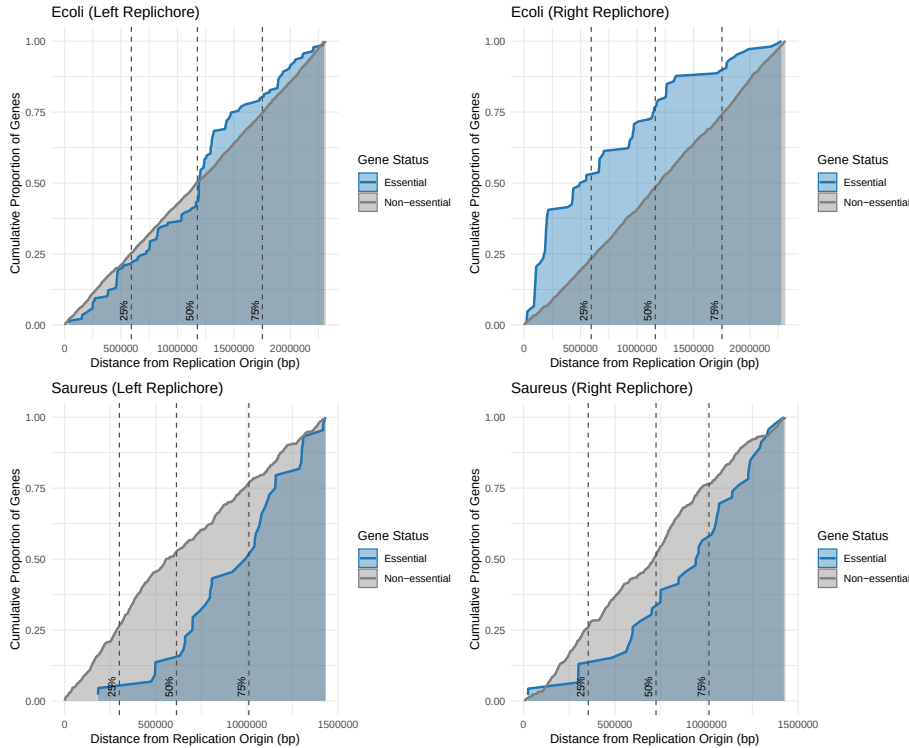


Figure 2.4. ECDF plot for each replichore. There is a sharp increase in the cumulative proportion of essential genes near the origin, especially in the right replichore of *ecoli* within the first 25% of the distance. In contrast, both *saureus* replichores tend to have essential genes reside further from the origin. The vertical dashed lines represent the 25th, median, and 75th percentiles of the distance from the origin for all genes.

2.4 Essential Gene Clustering

There are different definitions of clustering that can be applied to essential genes. A common definition is spatial clustering, where essential genes are considered clustered if they are located near each other, even if separated by occasional non-essential genes. A broader extension of this is to define essential regions, or sections of the genome consisting of a fixed gene count in which essential genes occur with higher probability than in non-essential regions. Viewed in this way, the genome can be understood as an alternating sequence of essential and non-essential regions, and this definition can be used to develop classifiers. A final, more restrictive definition is based on run lengths, where clustering refers to uninterrupted sequences of contiguous essential genes.

2.4.1 Spatial Clustering

The spatial clustering of essential genes is already evident in Figure 2.3. In the circular plots (Figures 2.1 and 2.2), there are dense clusters of essential genes, particularly around the origin for the *ecoli* genome. For the *saureus* dataset, there is evidence of clustering but this is closer to the terminus. The bulges present in the violin plots also clearly indicate a preference for essential genes to be located close to each other regardless of their position along the genome. One important point to note is that gene clustering in the *saureus* dataset is also present for the non-essential genes, which we note for both replichores but particularly for the left replichore.

2.4.2 Clustering by Contiguous Run Length

To investigate this mode of clustering, the distribution of contiguous run lengths of essential genes is examined within each replichore. A contiguous run is defined as a sequence of essential genes that appear consecutively without interruption from non-essential genes. Longer run lengths suggest local clustering of essential genes. To assess whether this clustering is significant, the observed run length distributions are compared to those generated under simulated random distributions of essential genes. This allows evaluation of whether the observed clustering arises by chance or reflects an underlying structure in the genomic data (Chapter 4). In the first instance, Figure 2.5 shows the run lengths of the essential genes for each of the replichores. The most common run length for all replichores is a single essential gene, which is expected due to the wide class imbalance. Run lengths are longer for the *ecoli* bacterium than the *saureus*, which may be partly attributable to the larger number of genes in the *ecoli* dataset. Alternatively, this pattern may indicate a stronger tendency for sequential clustering of essential genes in *ecoli*, as defined under this contiguous-run framework. In either case, while the observed run length distributions suggest the presence of clustering, a formal comparison with simulated data is necessary to determine whether this pattern deviates meaningfully from what would be expected under random placement of essential genes.

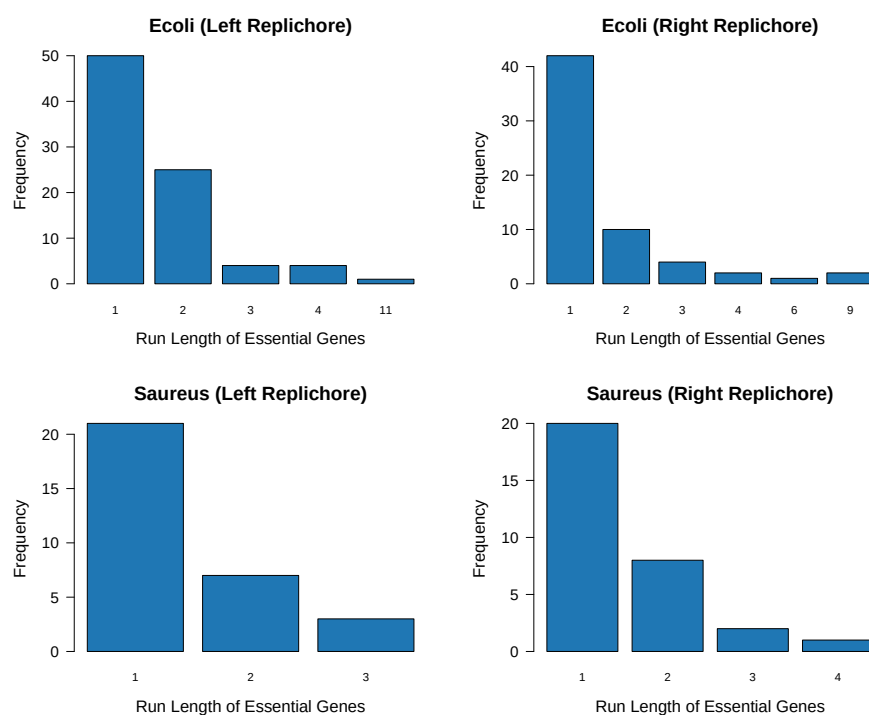


Figure 2.5. Histograms of the essential gene run lengths, split by replichore for each bacterium. We note a much greater frequency of essential genes of a run length one.

Chapter 3

Statistical Methods

This chapter presents the methodological framework of the report. The first element includes a description of Monte Carlo methods applied to permutation testing within a dataset. We use this methodology to determine clustering effects within our bacterium, in addition to their tendency to reside near the origin (Section 3.1). This is followed by a discussion of various techniques for classifying essential genes, with a critical evaluation of their suitability for the given data. Based on this assessment, Section 3.2.1 introduces Hidden Markov Models (HMMs) as an appropriate technique. The basic concepts are discussed, as well as its prevalence in gene sequence data. There is a brief overview of similar approaches to genome data within the literature, and how the current application is distinct. Section 3.3 provides an explanation of Bayesian Methods, with focus on their application to HMM's. Finally, Section 3.6 details the relevant performance metrics used to assess the classification accuracy of the HMM's and how the Bayesian framework can be used to compute these metrics.

3.1 Monte Carlo Methods

The broadest definition of Monte Carlo methods are a class of statistical methods that use repeated random sampling or numerical methods to give an estimate of a solution value (Auditore, L, et al, 2022). The general formulation is given below.

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and we wish to compute the expectation given by:

$$\mu = \mathbb{E}_X[f(X)] = \int_{\mathbb{R}^d} f(x)p(x)dx$$

where $p(x)$ is the probability density function over $\mathbb{R}^d$. A Monte Carlo estimator of μ using N independent samples $x_1, \dots, x_n$ is given by:

$$\hat{\mu}_N = \frac{1}{N} \sum_{i=1}^N f(x_i)$$

where $\hat{\mu}_N \rightarrow \mu$ as $N \rightarrow \infty$.

Permutation testing is an alternative method of significance testing that can be used with Monte Carlo simulations to estimate the null distribution of a test statistic, without relying on parametric assumptions. Permutation tests proceed in the following series of steps:

1. Calculating the test statistic of the observed data
2. Permuting or reshuffling the data based on a set of data labels
3. Calculating the test statistic on the newly permuted data
4. Repeating steps 2-3 over a set of simulations (typically 10000)
5. Comparing the observed test statistic to the null distribution of values generated from the permuted data.

This permutation removes any inherent order within the original dataset, which allows testing on whether any observed structure or association is stronger than what would be expected under the null distribution. If the observed value is as extreme or more extreme than 95% of the permuted statistics, we reject the null hypothesis at the 0.05 significance level (Kadiyala, E, 2025). This is a frequentist method used to determine if essential genes cluster, or if they reside near the replication origin.

In the context of permutation testing, the Monte Carlo permutation test result is of the form:

$$\hat{p} = \frac{1 + \sum_{i=1}^{M_p} \mathbb{I}(T^{(i)} \geq T_{obs})}{1 + M_p} \quad (3.1)$$

where: $\hat{p}$ is estimated p-value, M_p is the total number of random permutations, T_i is the value of the test statistic computed from the i th permuted dataset, $I()$ is the indicator function (1 if greater, 0 if not), T_{obs} is the test statistic computed from the original dataset. A continuity correction of 1 is added to prevent p-values of exactly 0.

3.2 Markov Chains

Markov Chains, introduced by Andrey Markov in 1906, are a type of stochastic process used to model systems that transition between states over time. We take X_k , $k \in \mathbb{Z}$ as a stochastic process where there is a countably finite number of values within a state space, S . The values of $\mathbb{Z}$ are positive. The process $\{X_k\}$ is Markovian if the probability of transitioning to the next state depends only on the present state and is independent of all previous states. Formally this can be expressed as:

$$P(X_{k+1} = j | X_k = i, X_{k-1} = i_{k-1}, \dots, X_1 = i_1) = P(X_{k+1} = j | X_k = i)$$

for all states $i_1, i_2, \dots, i_k, j \in S$ and all time steps $k \geq 1$.

Markov Chain are homogeneous if the transition probabilities do not change over time. The probability of moving from state i to state j at time k is the same as any other time step. This can be written as:

$$P(X_{k+1} = j | X_k = i) = P(X_2 = j | X_1 = i) = \gamma_{ij}$$

where γ_{ij} is the one-step transition probability from state i to state j . These transition probabilities satisfy the conditions $\gamma_{ij} \geq 0$ for all $i, j \in S$ and the sum of probabilities from any given state must equal 1:

$$\sum_{j=1}^m \gamma_{ij} = 1$$

where m is the possible number of states in S . A transition matrix can be defined with the probabilities as below:

$$\begin{bmatrix} \gamma_{11} & \gamma_{12} & \dots & \gamma_{1m} \\ \gamma_{21} & \gamma_{22} & \dots & \gamma_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{m1} & \gamma_{m2} & \dots & \gamma_{mm} \end{bmatrix}$$

where Γ is a matrix of size $(m \times m)$, where m denotes the number of states in the Markov chain (Mirzai, P, 2019).

3.2.1 Hidden Markov Models

A Hidden Markov model (HMM) is a statistical model used when the system being studied evolves over time (or position) through a sequence of hidden states, but produces observable outputs that are dependent on those hidden states. HMMs are useful in cases where the underlying structure of the process cannot be observed directly, but indirect observations provide information about it. HMM's are widely used in applications such as speech recognition, but are well-suited to modelling biological sequences due to their ability to work with sequential data and with a set of unknown hidden states representing the biological phenomena (Yoon, B, 2009).

The hidden state sequence may be denoted by $Z = \{Z_1, \dots, Z_T\}$ where each $Z_t \in S$ represents the hidden state at time/position t . The observed sequence is $Y = \{Y_1, \dots, Y_T\}$ where each $Y_t \in O$ is an observed variable emitted at time t . Y is linked to the Markov chain with Z governing the distribution of Y . As

with Markov Chains, the hidden state sequence satisfies the Markov property meaning the probability of transitioning to the next hidden state depends only on the current state:

$$P(Z_{t+1}|Z_t, Z_{t-1}, \dots, Z_1) = P(Z_{t+1}|Z_t)$$

The observed data at any time point depends only on the corresponding hidden state and is conditionally independent of past observations and states:

$$P(Y_t|Y_{t-1}, \dots, Y_1; Z_t, \dots, Z_1) = P(Y_t|Z_t)$$

The emission probability is the probability of observing a particular Y_t given the hidden state Z_t :

$$e_z(y) = P(Y_t = y|Z_t = z)$$

Depending on the nature of observations, the emission probability will take a probability mass function or density function over the set of possible observations (Mirzai, P, 2019). The final component to completely specify a HMM is the initial state distribution, or how likely the system is to start in each possible hidden state. This is denoted by:

$$\pi_i = P(Z_1 = i), \quad \text{for } i \in S$$

This gives the model a starting point before any transitions or emissions occur. The three probability measures π_i , $e_z(y)$, and the transition probabilities γ_{ij} specify the HMM (Yoon, B, 2009).

3.3 Bayesian Modelling

3.3.1 Bayesian Theory

We use a Bayesian statistical framework for the application of HMM's. Bayesian inference provides a structured approach to updating beliefs about unknown model parameters using observed data. A parameter θ represents a set of unknown model parameters whose values are to be estimated. When data x is observed, beliefs about θ are updated using the posterior distribution, denoted by $\theta|x$. This posterior represents the conditional probability of the parameter given the data and incorporates all information relevant to θ after observing $X = x$ (Lambert, B, 2018). This is Bayes Theorem:

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)} \propto p(x|\theta)p(\theta)$$

where: $p(\theta|x)$ is the posterior of θ given x , $p(\theta)$ is the prior distribution of the parameters (representing the initial belief) and $p(x|\theta)$ represents the likelihood which is the probability of the distribution given data x . The sum of the posterior is $\sum_{\theta} p(x|\theta)p(\theta)$ when θ is discrete, and $\int p(x|\theta)p(\theta)d\theta$ when continuous. The normalizing constant $p(x)$ ensures the posterior is a valid probability distribution. However, calculating $p(x)$ often involves complex or high-dimensional integrals that are intractable, especially in models like Hidden Markov Models with hidden variables. Computational techniques such as Markov Chain Monte Carlo (MCMC) methods are used to address this issue (Lambert, B, 2018).

3.3.2 Bayesian Statistics and MCMC

MCMC algorithms generate samples from the posterior distribution without requiring explicit calculation of $p(x)$. These samples can then be used to approximate posterior summaries and make inference feasible in complex Bayesian models. MCMC algorithms generate a sequence of samples where each sample depends on the previous one. This forms a Markov chain that over time converges to the target posterior distribution. At the start of the Markov Chain, the initial samples are not representative of the true posterior distribution and these are discarded (known as ‘burn-in’). The MCMC approach used in this report is the Hamiltonian Monte Carlo method (Lambert, B, 2018).

3.3.3 Hierarchical Bayesian Models

A hierarchical Bayesian model extends the simple Bayesian analysis, where the prior $p(\theta)$ and likelihood $p(x|\theta)$ are used to compute a posterior probability $p(\theta|x)$, to include uncertainty in the prior itself by allowing it to depend on additional unknown hyperparameters (e.g. V, W). In this framework, the prior becomes conditional $p(\theta|V, W)$ from which higher level priors $p(V)$ and $p(W)$ need to be specified for the hyperparameters. This leads an extended posterior of the form:

$$P(\theta, V, W|x) \propto p(x|\theta)p(\theta|V, W)p(V)p(W)$$

The process can continue so that the parameters V and W can have dependence on an additional parameter Z and so on until dependencies cease. This hierarchical structure allows information to be shared across groups and improves estimation when data is sparse. In the context of essential gene classification, hierarchical Bayes is well-suited because they naturally account for different sources of variation both at the global state level and at the local genomic region level (Li, G and Shi, J, 2012).

3.4 Bayesian HMM

In a Bayesian framework, the parameters of the HMM and the hidden states themselves are treated as random variables. The aim of Bayesian inference is to estimate the joint posterior distribution over both the hidden state sequence $Z = Z_1, \dots, Z_T$ and the model parameters θ given an observed sequence $Y = Y_1, \dots, Y_T$.

The posterior distribution is expressed as:

$$P(Z, \theta | Y) \propto P(Y | Z, \theta) P(\theta)$$

where $Z = Z_1, \dots, Z_T$ is the unobserved sequence of hidden states, each $Z_t \in S$. $Y = Y_1, \dots, Y_T$ is the observed sequence, with each $Y_t \in O$. θ represents all model parameters, including the transition probabilities $\Gamma = \gamma_{ij}$ and emission parameters $\xi = \xi_i$ (e.g. probabilities or parameters governing the distribution of Y_t given $Z_t = i$) (Ghahramani, Z, 2001).

3.5 Model Selection Criteria

In this report, we apply Markov Chain Monte Carlo (MCMC) methods to perform Bayesian inference to develop a HMM for the datasets. There are a couple of metrics used to determine if the algorithm has converged to the target posterior distribution. Convergence diagnostics evaluate the reliability of the generated samples. The key diagnostics are the Gelman–Rubin statistic ($\hat{R}$) and the Effective Sample Size (N_{eff}).

The Gelman-Rubin statistic ($\hat{R}$ or Rhat) is a convergence heuristic which calculates the ratio of the between-chain variance and the within-chain variance given by:

$$\hat{R} = \sqrt{\frac{\hat{V}}{W_{wc}}} = \sqrt{\frac{\frac{L-1}{L} W_{wc} + \frac{1}{L} B}{W_{wc}}} \quad (3.2)$$

where $B = \frac{L}{J-1} \sum_{j=1}^J (\bar{x}_j - \bar{x})^2$, j is the chain, L is the length of the chain, $\bar{x}$ is the overall mean across chains, $\bar{x}_j$ is the mean of the j th chain. W_{wc} is the average within-chain variance $W_{wc} = \frac{1}{J} \sum_{j=1}^J s_j^2$. If all chains have converged to the same posterior distribution, their within-chain and between-chain variances should be similar. If the $\hat{R}$ values are close to 1 (< 1.05), then the chains of the model have mixed well and converged to the true distribution (Peng, R, 2022).

In MCMC, samples are generated sequentially which can lead to autocorrelation where each sample is partially dependent on the previous one. The Effective Sample Size (ESS) indicates how much independent information is lost due to correlation in the MCMC samples. It is given by:

$$N_{\text{eff}} = \frac{N_s}{\sum_{-\infty}^{\infty} \rho_t} \quad (3.3)$$

where N_{eff} is the net effective sample size, N_s is the number of samples. ρ_t is the autocorrelation given by:

$$\rho_t = \frac{\text{cov}(X_i, X_{i+t})}{\sigma_{X_i} \sigma_{X_{i+t}}}$$

where σ_X is the standard deviation of X for a given lag of time t . A lower effective sample size increases the uncertainty of estimated values such as the posterior means or variances. If the samples in the MCMC chain were independent, the number of samples would be the same as the effective sample size. The effective sample size will typically be smaller than the total number of iterations, indicating that some of the samples are effectively redundant. Ideally, the effective sample size is large and closer to the number of samples generated by the chains (Stan Development Team, 2014).

A final metric used to assess region size is Shannon entropy, which measures uncertainty in the inferred state probabilities. In the context of the dissertation, higher entropy indicates that the model is less confident in classifying a region as essential or non-essential. This metric is used to assess the optimal region size for the model (Saraiva, P, 2023). The equation is of the form:

$$H(p) = -p \log_2(p) - (1 - p) \log_2(1 - p) \quad (3.4)$$

Where: p = probability of essential gene or region, $1 - p$ = probability of non-essential gene or region.

3.6 Performance Metrics

3.6.1 Classification

The performance of the classifier can be understood from a set of performance metrics that allow benchmarking and comparison of the HMM. Posterior probabilities are extracted from the model and averaged across the MCMC draws, which gives the mean posterior probability per region. These can be converted into class predictions where $\hat{y}_g \in \{0, 1\}$ is the predicted label (1 = essential) for gene g , $y_g \in \{0, 1\}$ is the true label. The confusion matrix is given by:

- True Positives (TP): $\hat{y}_g = 1, y_g = 1$
- True Negatives (TN): $\hat{y}_g = 0, y_g = 0$
- False Positives (FP): $\hat{y}_g = 1, y_g = 0$
- False Negatives (FN): $\hat{y}_g = 0, y_g = 1$

Here true positives are the number of essential genes correctly identified and true negatives are correct predictions of non-essential genes. False positives are non essential genes incorrectly predicted as essential, and false negatives are essential genes that are incorrectly predicted as non essential. We can use these outputs to calculate the true positive rate (sensitivity) and the true negative rate (specificity) (Hicks, S. A. et al, 2022). These are given by:

$$\text{True Positive Rate (TPR)} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{True Negative Rate (TNR)} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

For gene classification, sensitivity measures the model’s ability to correctly identify essential genes and a higher sensitivity indicates that the model effectively detects most essential genes, minimizing false negatives. Specificity measures the model’s ability to correctly identify non-essential genes and a higher specificity means fewer false positives.

3.6.2 ROC-AUC

The classifier must minimize both the false positive and false negative rates to ensure reliable classification. Receiver Operating Characteristic (ROC) curves provide a visual representation of this trade-off by plotting the true positive rate (sensitivity) against the false positive rate (1 - specificity) at various threshold settings. The Area Under the ROC Curve (ROC-AUC) summarizes the classifier’s overall ability to discriminate between essential and non-essential genes. This is given by:

$$\text{AUC} = \int_0^1 \text{TPR}(t) \, d\text{FPR}(t)$$

where: FPR is the False Positive Rate. For this report, the AUC measures the probability that a randomly chosen essential gene is assigned a higher posterior probability than a randomly chosen non-essential gene. An AUC of 1 indicates perfect separation, while an AUC of 0.5 suggests only random chance of separation (Totsch, N and Hoffmann, D, 2021). There are concerns that there is potential inflation of ROC-AUC scores in the presence of substantial class imbalance, though studies have demonstrated that the ROC-AUC metric is invariant to class proportions and remains robust under such conditions (Richardson, E, et al, 2024). Despite the imbalance in the TraDIS datasets, where essential genes constitute a small minority, ROC-AUC is an appropriate metric for evaluating performance. Accuracy is not included due to the wide class imbalance. A classifier that labels all genes as non-essential would achieve a high accuracy by merely assigning all predictions to the majority class.

3.7 Classification Approaches

One of the secondary aims of the dissertation is to classify essential genes based on the genomic features of the datasets. This is explored in Chapter 4 leveraging the gene clustering, distance from the replication origin and insertion density features. Machine Learning techniques are well suited to classification problems, and have been commonly used for essential gene classification (Zhang, X, et al, 2016). Techniques employed include Support Vector Machines (SVM), Random Forests (RF), K-Nearest Neighbors (KNN) and Artificial Neural Networks (ANN). SVM is a supervised machine learning algorithm that aims to find the best boundary (hyperplane) that separates data points of different classes with the widest possible margin (Bansal, M. et al, 2022).

This approach is not ideally suited to TraDIS data as SVM tends to perform poorly on highly imbalanced datasets with a small minority positive class (e.g. with a small proportion of essential genes). Additionally, the datasets are not linearly separable due to overlapping patterns and its sequential nature, making it challenging for SVMs to define a clear boundary without significant complexity (Bansal, M. et al, 2022). RF's are ensemble learning methods that create a large number of decision trees during training. Each tree is built from a random subset of the data and features. The final classification is made by aggregating the predictions of all trees, typically through majority voting (Aria, M, et al, 2021). This technique is a viable approach for the classification, but are limited by their computational cost which scales very quickly with additional features as well as being highly complex and lacking intuitive interpretability of the biological phenomena.

KNN's are a non-parametric, supervised learning algorithm that classifies the label of a data point based on the majority class among its closest neighbors in the feature space. While KNNs are simple to implement and easily interpretable, it faces scalability challenges due to its high computational cost and sensitivity to the choice of k and the distance metric, which must be specified. Additionally, KNNs do not provide a direct probabilistic output, limiting its ability to provide uncertainty in classifications, which can be a disadvantage in the context of biological classification (Bansal, M. et al, 2022).

ANN's are a set of computational models composed of interconnected units called neurons, organized into layers (an input layer, one or more hidden layers, and an output layer). Each neuron receives inputs, applies a weighted sum, passes it through a non-linear activation function, and produces an output. As with other techniques, ANN's have been applied extensively for essential gene classification (Zhang, X, et al, 2016) by using biological features of the essential genes to perform the separation. The limitations of ANN's include their complexity and computational expense, as well as their requirement for careful feature selection and engineering which can lead to overfitting. They typically require large training data sets to perform well and, like RF's, are less interpretable for biological insight (Koumakis, L, 2020).

3.8 Literature Review

3.8.1 ESSENTIALS

Multiple software packages have been developed to predict essential genes from TIS data. The ESSENTIALS algorithm is designed to analyze TIS data by processing read counts mapped to genomic insertion sites. A read is a small section of DNA sequence that maps to the insertion site and a read count is the number of sequencing reads that map to an insertion site. A single insertion site can generate multiple reads. For the ESSENTIALS algorithm, all possible transposon insertion sites on the genome are identified, and the surrounding DNA sequences are collected. The raw sequencing reads are then converted into a readable format, sorted by barcodes using a flexible matching method, and filtered to keep only those containing transposon sequences. These filtered reads are matched to the insertion sites. LOESS regression is used to mitigate any bias caused by the location on the genome or differences in read coverage. This software assumes the read counts are negatively binomially distributed. Testing finds genes with significantly fewer insertions are likely essential, which matches the expected biological behaviour (Zomer, A, et al, 2012). A limitation of this algorithm is that it tends to over predict non-essential genes as essential genes (false positives) when using its reported p-values for classification, and is at risk of classifying essential genes as non-essential (false negatives) when insertions are made at the N- or C- termini (beginning and end of a gene) (Ioerger, T, 2022).

3.8.2 TRANSIT and Hidden Markov Models

TRANSIT is a Python based software package used for the analysis of Tn-Seq data. This dataset is generated from mariner transposons that insert at thymine-adenine (TA) dinucleotide sequences, which differs from the Tn5 transposons which insert randomly in a genome (datasets in this dissertation). The mariner insertions are limited to discrete, defined positions along the genome. The software analyses sequence data through two methods, Gumbel Analysis and HMMs. The Gumbel analysis detects essential genes by identifying long consecutive sequences of TA sites without insertions since these gaps are unlikely to occur by chance in non-essential regions. This method uses the Extreme Value Distribution to assign posterior probabilities, categorizing genes as essential, non-essential, or uncertain. Gumbel analysis is tolerant of insertions near gene ends but performs less well with less-saturated libraries and shorter genes. It can also merge data from multiple replicates to improve accuracy (Ioerger, T, 2022).

The HMM approach models each TA site probabilistically, considering both the insertion counts and neighboring sites. This algorithm classifies genes into one of four states: essential, growth-defect, non-essential, and growth-advantaged depending on the varying quantity of insertions. They are the q_{ES} (essential)

state, where insertions are not tolerated and expected to have zero or near-zero reads. The q_{GD} (growth-defect) state where insertions impair growth but are not lethal and are characterized by low read counts. The final two are q_{NE} (non-essential) where insertions are tolerated and have a wide range of read counts and q_{GA} (growth-advantage) where disruption may enhance growth and often show higher-than-average read counts. The likelihood of observing a specific read count at a TA site is modeled in the study by using a geometric distribution (emission probabilities), as in Tn-Seq experiments sites with lower counts are more common and sites with high counts are more rare.

$$p(c_i|q_{NE}; \theta) = (1 - \theta)^{c_i} \theta$$

where θ is the Bernoulli probability of insertion for the geometric distributions. The MLE for the parameter θ is $\frac{1}{\bar{c}}$ where $\bar{c}$ is the mean read count at inserted TA sites. Geometric distributions are also used for the likelihood functions of other states. For $\theta_{ES} = 0.99$ as these sites are essential so tolerate 0 or near 0 insertions. For the other θ we have:

$$\frac{1}{(0.01x\bar{c} + 2)}$$

$$\frac{1}{5\bar{c}}$$

The transition probabilities in this model were used to capture dependencies between neighboring insertion sites. The assumption of the model is that the probability of remaining in the same state should be nearly 1 for all states T_{aa} , while the off-diagonal elements in the transition matrix T_{ab} may be close to 0 for $a \neq b$ so the state transition probabilities are:

$$T_{ab} = p(q_{i+1} = b | q_i = a)$$

They then determined the insertion probability estimates (P_{ins}) by discarding regions with ≥ 10 insertion sites and calculating the remaining regions insertion density. They determine the minimum length of an essential region (r^*) as the smallest run where the geometric probability is less than 0.01. The transition probability of remaining in the same region is given by:

$$T_{aa} = 1 - (\lambda_{NE}(0))^{T^*}$$

where $\lambda_{NE}(0)$ is the likelihood of zero insertions in a non-essential region. The model developed in this analysis differs in that it uses a two state Bayesian HMM on smoothed regional data (DeJesus, M and Ioerger, T, 2013).

Another study specified a HMM framework for mariner-based TnSeq data for the bacterium *Vibrio cholerae* by modeling TA sites as emissions and categorizing hidden states as essential, neutral, or sick. The continuous read counts at each TA site were binned into seven levels based on their distribution across the genome. This binning was completed to distinguish known sick genes from essential ones. The model was trained using manually annotated regions and statistical signals from the genome, allowing classification of each TA site into one of the three states. Essential genes were defined as those whose coding regions contained only TA sites classified as essential. Regions that had mixed classifications but included a stretch of essential sites designated as domain essential, while loci with predominantly sick TA sites were labeled sick. All other regions defaulted to neutral. To determine the confidence for each HMM assignment, the posterior probabilities for every TA site was assessed after being categorized. By analyzing the frequency and depth of transposon insertions at TA sites, the HMM was able to detect “sick” genes distinguishing them from growth-neutral loci that are missed by traditional sliding window (SW) or binary insertion-based methods. Experimental validation confirmed that mutants in genes with very low average read counts (1–2 reads/insertion) displayed impaired growth, supporting the accuracy of HMM predictions. However, the precision decreased for genes with higher read counts, indicating potential false positives (Chao, M, et al, 2013).

Chapter 4

Results and Interpretation

This chapter applies the methods described in Chapter 3 to the datasets. The first part conducts hypothesis tests on the original and simulated data to assess if essential genes show a positional bias toward the replication origin. It is also applied to assess if there is evidence of essential gene clustering (Section 4.1.2). The test statistics compare mean and median cluster sizes, as well as the mean and median distances of essential genes from the replication origin, between the observed (real data provided for the dissertation) and simulated datasets. The second part of the chapter focuses on the development of an HMM to classify genomic regions as essential or non-essential. An initial baseline model is constructed by counting essential genes within a region without consideration to insertion density or preference to the origin (Section 4.2.1). This serves as a benchmark for evaluating classification performance. The model is then extended by incorporating insertion density (Section 4.4.1) and distance to the replication origin (Section 4.5) into separate Hierarchical Bayesian HMM's. This is done by modifying the emission probabilities and prior distributions within the HMM. The effectiveness of the classification is assessed through a range of metrics, primarily AUC (Section 3.6.2), in addition to the stability of the model. The final part of the chapter simulates realistic genomic data based on the features of datasets (Section 4.6). By varying features such as the proportion of essential genes, these simulations provide controlled experiments to examine the influence on the classification performance in addition to the model robustness when presented with new data.

4.1 Simulation and Permutation Testing

4.1.1 Distance from the Replication Origin

Permutation testing is used to compare the distribution of essential genes within the genome. This is done by randomly permuting essential gene labels from the observed data while preserving the proportions of essential and non-essential genes. This removes the inherent biological structure of the observed data, from

which the significance of the difference using appropriate test statistics can be determined. The original approach for this problem used a chi-squared test. The genome would be partitioned into q bins (e.g. deciles), and the observed counts of essential genes per bin compared with the expected counts under a uniform random distribution. However, this approach was discarded as it required grouping the continuous data into arbitrary bins, which can introduce bias depending on binning choices and possibly lead to erroneous results. The hypothesis for the test is therefore evaluated using the following framework, where each permutation simulates a scenario under the null hypothesis:

- Null hypothesis (H_0): The locations of essential genes are randomly distributed along the genome and do not tend to reside near the replication origin.
- Alternative hypothesis (H_1): The essential genes are non-randomly distributed, with the likelihood that they are located closer near the replication origin.

The test statistics used for the simulation are the mean and median distances of the essential genes from the replication origin. Let the distances of the essential genes from the replication origin be denoted by $d_1, d_2, \dots, d_{n_e}$, where n_e is the number of essential genes in a given replicore. $\bar{d}_e$ is then the sample mean distance from the replication origin for essential genes. The median distance is the middle value of the distances. The p-value for the mean/median distances is calculated by comparing the observed mean/median distance from the replication origin to the distribution of simulated mean/median distances. We count the number of simulations where the randomly generated mean/median distance is less than or equal to the observed value. This count is then divided by the total number of simulations to obtain the proportion of simulations that would yield a mean/median distance as extreme as or more extreme than the observed one. This is an applied form of Eq (3.1), where: M = total number of simulations, T_{obs} observed test statistic (e.g. mean distance of essential genes from the replication origin), T_i = test statistic from the i th simulation (e.g. mean or median simulated distance of essential genes from the replication origin), $\mathbb{I}(\cdot)$ = indicator function, which equals 1 if the condition is true and 0 otherwise.

For the simulation, the number of essential and non-essential genes for each replicore is counted and stored. The number of simulations chosen for this procedure is 10000. The simulation then randomly permutes the essential and non-essential gene labels to create a new random sequence of genes. The mean and median distance from the origin for all the genes labeled as essential in the shuffled dataset are calculated. It stores this mean (or median) in an array at the index i , looping 10000 times to give a distribution of new mean (or median) distances.

If the p-value is very low (e.g. < 0.01), it suggests that the distance from the replication origin for the essential genes is unlikely to have occurred by random chance, supporting the idea that essential genes are non-randomly positioned

Table 4.1. Mean and Median p-values for Replication Origin Proximity by Replichore for Ecoli and Saureus.

Dataset	Replichore	Replication Origin Proximity p-values	
		Mean Distance p-value	Median Distance p-value
Ecoli	Left	0.42166	0.57714
Ecoli	Right	0.00010	0.00010
Saureus	Left	1.00000	0.99990
Saureus	Right	0.99940	0.99970

near the replication origin. Table 4.1 shows the calculated p-values for the test statistics split by bacterium and replichore. The right replichore for the ecoli bacterium has a value of 0.00010 for both mean and median distance, which suggests essential genes on the right replichore are preferentially located at the origin. For this replichore, we may reject the null hypothesis that the essential genes are randomly distributed along the genome. However, for the left replichore, the p values are ~ 0.41 and ~ 0.59 for the mean and median distances, respectively. This suggests the observed positions of essential genes are not significantly different from random placement. This is in agreement with the exploratory data analysis performed in Section 2.3, which did not highlight a tendency towards the origin.

For the saureus bacterium, the p values of very close to 1 (rounded to 1 by the limit of resolution) regardless of replichore or test statistic. This suggests that there is no evidence of essential genes preferring the origin and the observed distributions are entirely consistent with what would be expected under random placement. This agrees with the prior data analysis where we noted that the essential genes tend to reside further away from the origin.

Figure 4.1 shows histograms of the simulated mean and median distances from the origin for the ecoli essential genes, split by replichore (left and right). The observed distances from the original data are overlaid to highlight how they compare to the distribution under random assignment. For the right replichore, the observed distance is substantially lower than any of the simulated values. In contrast, the observed distance of the left replichore falls well within the range of simulated values implying no strong deviation from random placement. The values are close to the mode, indicating that these are typical values under the null distribution. Figure 4.2 displays the corresponding histograms for saureus, showing the simulated distributions of mean and median distances from the origin for each replichore. In both the left and right replichores, the observed distances are above virtually all the simulated values. This indicates that, under random placement, essential genes are expected to be positioned closer to the origin than they are in the observed genome. While the observed distances are always greater than those from the simulated distributions, the gap is not as wide as in the right replichore of ecoli. Some values at the upper tail of the

simulated distributions approach the observed distances, indicating that these results are not as strongly divergent from a random sample as in *ecoli*.

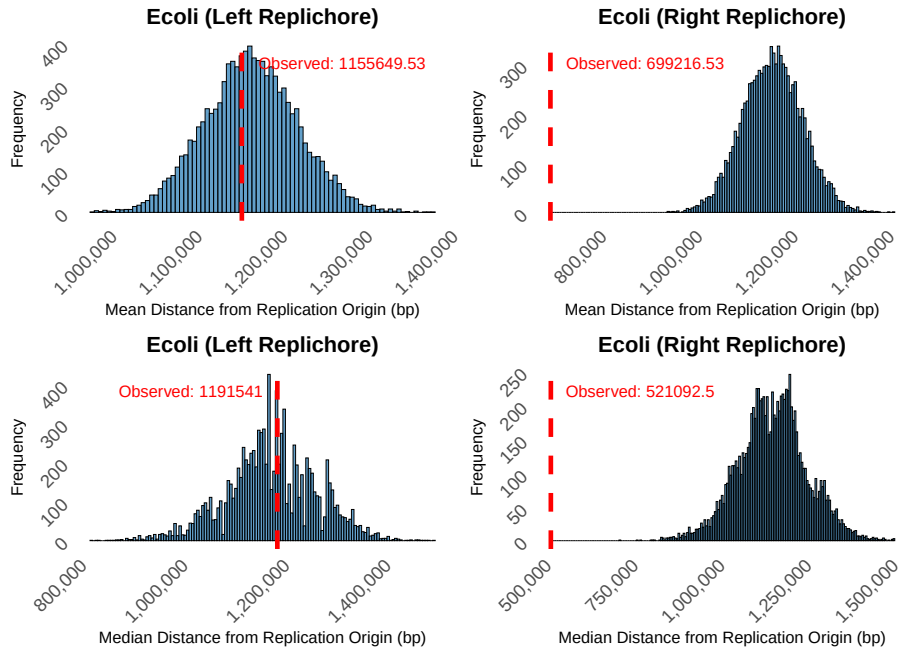


Figure 4.1. Histogram plot of simulated of the mean and median distances from the replication origin for the *ecoli* dataset. The observed mean/median distance is given by the red dashed line.

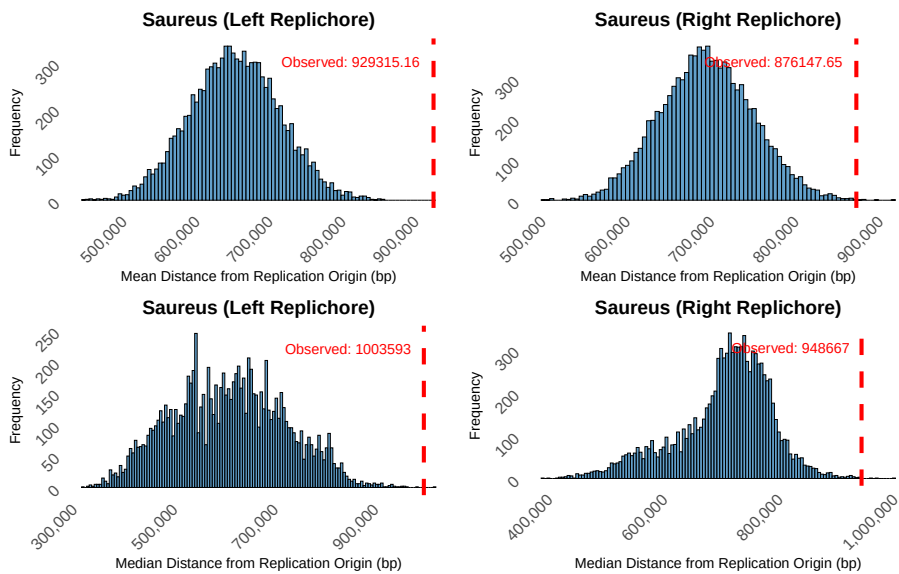


Figure 4.2. Histogram plot of simulated mean and median distances from the replication origin for the *saureus* dataset. The observed mean/median distance is given by the red dashed line.

4.1.2 Essential Gene Clustering

A similar approach of simulation and permutation testing can be used to investigate essential gene clustering. In this case, it is used to assess whether essential genes tend to form clusters by comparing clustering test statistics from the observed data to those generated under random permutation. The hypothesis are as follows:

- Null hypothesis (H_0): Essential genes are randomly distributed along the genome. Any observed clustering (contiguous runs of essential genes) is no greater than would be expected by chance.
- Alternative hypothesis (H_1): Essential genes are non-randomly distributed along the genome, forming clusters (longer or more frequent contiguous runs of essential genes) that are unlikely to occur under a random distribution.

The test statistics are the mean and maximum cluster sizes of the observed and simulated data. A cluster is defined previously as a contiguous sequence of essential genes labels in the genome. Let $L = (l_1, l_2, \dots, l_n)$ where $l_i \in (0, 1)$, denote the binary sequence of essentiality labels for the n genes along the genome. $l_i = 1$ denotes an essential gene and $l_i = 0$ denotes a non-essential gene. For a given sequence L we have:

$$C = \{c_1, c_2, \dots, c_m\}$$

where C is a set of cluster sizes and where each c_j is the length of a contiguous subsequence of 1s in L . The mean cluster size is then given by:

$$\bar{c} = \frac{1}{m} \sum_{j=1}^m c_j$$

where: m is the number of clusters. The maximum cluster size is given by:

$$c_{\max} = \max_{1 \leq j \leq m} c_j$$

As before, we generate 10,000 random permutations of the essential/non-essential gene labels, preserving the total number of essential genes. Table 4.2 shows the p-values from permutation tests for each replicore, using both the mean and maximum cluster sizes of essential genes. For the mean cluster sizes, all replicores have p-values significantly less than 0.01, meaning that in none of the simulations did the mean cluster size exceed the observed value. This strongly suggests that essential genes are more clustered in the actual data than would be expected under a random distribution. For maximum cluster size, both *ecoli* replicores also show p-values of 0.0001, indicating that the largest observed essential gene clusters in *ecoli* are highly unlikely to be due to

Table 4.2. Mean and Maximum Cluster p-values by Replichore for Ecoli and Saureus.

Dataset	Replichore	Cluster p-values	
		Mean Cluster p-value	Max Cluster p-value
Ecoli	Left	0.0001	0.00010
Ecoli	Right	0.0001	0.00010
Saureus	Left	0.0001	0.15488
Saureus	Right	0.0001	0.01070

chance. In contrast, the saureus replichores have higher p-values (0.151 for the left, 0.0121 for the right), suggesting that large clusters could plausibly occur by random chance. However, these values are still low indicating that some degree of clustering is present in the saureus genome that cannot be replicated from random distributions.

Figure 4.3 shows the distribution of mean cluster sizes for the simulated data, split by replicore and genome. In each plot, the observed mean cluster size is marked by a red dotted line. The observed values lie far to the right of the simulated distributions, with none of the simulations approaching the observed mean cluster sizes, even in the extreme tails. This supports the test statistic results, in that the mean cluster size of essential genes in all replichores is larger than by chance. The max cluster size histograms are plotted in Figure 4.4. In this case, the ecoli genome has an observed maximum cluster size far above all simulated cluster sizes for both replichores (11 and 9 for the left and right, respectively). This shows that ecoli has at least one large cluster of essential genes that is unlikely to occur randomly. In contrast, the saureus genome shows maximum cluster sizes (3 and 4 for the left and right replichores) that are close to the simulated distributions, suggesting that clustering in this genome is consistent with a random distribution. This highlights a difference between the genomes, where the ecoli bacterium shows a high degree of clustering with many long contiguous clusters, while clustering in the saureus genome is diffuse with many smaller, less tightly clustered groups.

4.2 Hidden Markov Models

4.2.1 Region Based Classification

A Hidden Markov Model (HMM) is constructed by grouping genes into contiguous regions of fixed size and modeling the number of essential genes in each region (Section 3.2.1). This regional approach offers several advantages over a classifier based on individual genes. First, gene essentiality can vary depending on growth conditions and bacterial strain, and gene-level data is noisy due to spurious insertions that may incorrectly label non-essential genes as essential. Aggregating

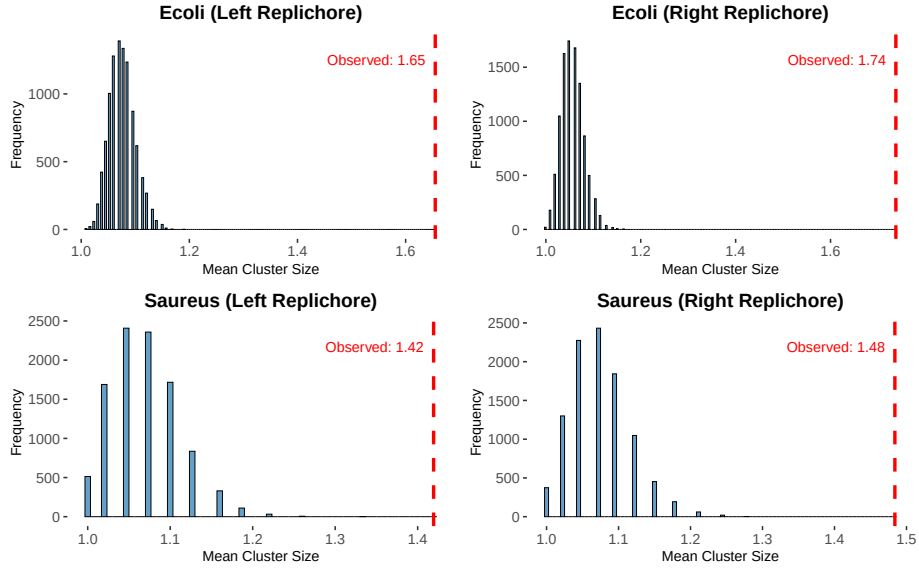


Figure 4.3. Histograms of simulated mean cluster sizes for the *ecoli* and *saureus* datasets. The observed mean cluster size is given by the red dashed line. There is a wide gap between the simulated values and the observed value.

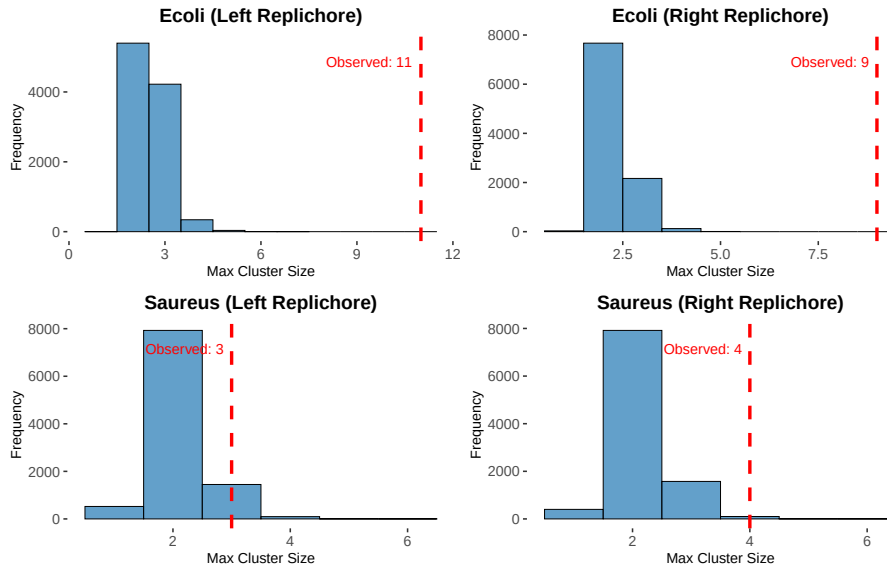


Figure 4.4. Histograms of simulated maximum cluster sizes for the *ecoli* and *saureus* datasets. The observed maximum cluster size is given by the red dashed line. There is a wide difference between the simulated max cluster size and the observed max cluster size for the *ecoli* data. The *saureus* dataset does not exhibit large cluster sizes, and the observed cluster size is within typical values.

over the regions reduces the influence of this noise and obtains more stable classifications. Secondly, using regional HMM's imposes a Markovian dependence

structure across the regions which makes interpretation of the transitions between essential and non-essential regions simple. This region HMM, which does not include the insertion densities or distance from the origin, is termed the “Basic” HMM. This is developed as a benchmark to compare against more complex models including the additional covariates.

Gene level models were attempted at an initial development phase, but required intense computational resources and time. Each replicore contains several thousand genes (e.g. *ecoli* ~ 4000 genes, *saureus* ~ 1400 genes; Table 2.2), and the HMM forward-backward algorithm scales as $O(n \cdot H^2)$ where n is the number of genes and H is the number of hidden states. Covariates like insertion density or distance from the origin introduce tens of thousands of hierarchical parameters (per state). We apply the current model without consideration to the gene level insertion densities and distance from the replication origin to develop a base case for HMM with regional classification for the replicores. This base model is expanded in (Section 4.3) to include these covariates.

4.2.2 Model Development

The genome is partitioned into N_r contiguous regions, each containing n genes. For each region $r = 1, \dots, N_r$, we observe the number of essential genes y_r , which is modeled as a binomial random variable. This is appropriate since y_r represents the number of successes (essential genes) out of a fixed number of trials (n genes per region). The hidden state z_r indicates whether the region is essential ($z_r = 2$) or non-essential ($z_r = 1$). It is assumed that the hidden states follow a first order Markov Chain, which is that the probability of transitioning to a new state only depends on the current state. Our initial distribution is given by:

$$P(z_1 = i) = \delta_i, \quad i \in \{1, 2\}, \quad \text{with } \delta_1 + \delta_2 = 1$$

where δ_i is the probability that the first region begins in state i . Specifically, $\delta_1 = P(z_1 = 1)$ is the probability that the first region is non-essential, and $\delta_2 = P(z_1 = 2)$ is the probability that it is essential. The model uses a Dirichlet prior [5,5] for the initial distribution, which reflects the assumption of equal probability of starting in either an essential region or a non-essential one. The choice of [5,5] rather than a flat prior such as [1,1] reflects a weakly informative belief that encodes moderate confidence in a balance between the two states while still allowing the data to dominate if strong evidence arises. A Dirichlet distribution is a generalization of the Beta distribution and the multinomial generalization of a binomial distribution. It is used for modeling the transition probabilities in the HMM because it moves between discrete states that are best modelled by a categorical or multinomial distribution (Bishop, C, 2006). The probability density function of the Dirichlet distribution is given by:

$$f(\theta; \alpha) = \frac{1}{B(\alpha)} \prod_{i=1}^K \theta_i^{\alpha_i-1}$$

where: θ = probability parameters of the categorical distribution, K = number of parameters or states, $\theta = (\theta_1, \theta_2, \dots, \theta_K)$ such that $\sum_{i=1}^K \theta_i = 1$ and $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_K)$ are the concentration parameters, with $\alpha_i > 0$. $B(\alpha)$ is a multivariate Beta function defined by:

$$B(\alpha) = \prod_{i=1}^K \frac{\Gamma(\alpha_i)}{\Gamma(\sum_{i=1}^K \alpha_i)}$$

where Γ is the Gamma function. For the transition probabilities, these are defined based on the Markov chain as:

$$P(z_{n+1} = j | z_n = i) = \gamma_{i,j}, \quad i, j \in \{1, 2\}$$

where $\gamma_{i,j}$ is the probability of transitioning from state i to state j . These transition probabilities are collated in a transition matrix given by:

$$\Gamma = \begin{bmatrix} \gamma_{1,1} & \gamma_{1,2} \\ \gamma_{2,1} & \gamma_{2,2} \end{bmatrix}$$

where: $\gamma_{1,1}$ probability of staying in a non-essential region, $\gamma_{1,2}$ probability of moving from a non-essential region to an essential region, $\gamma_{2,1}$ probability of moving from an essential region to a non-essential region, $\gamma_{2,2}$ probability of staying in an essential region. In our model, we place Dirichlet priors (Dirichlet[2,2]) on each row of Γ , where the first and second rows correspond to transitions from non-essential and essential states, respectively. This prior is weakly informative as it has small equal values (2 and 2) which don't strongly favor one transition over another. In the context of essential gene regions, it allows the model to capture clustering of essential genes and switches between essential and non-essential regions but also allowing the data to determine the transition probabilities. The emission distribution can be modeled as the observed number of essential genes in a region given the hidden state:

$$Y_r | Z_r = i_r \sim \text{Binomial}(n, p_i), \quad i \in \{1, 2\}$$

where Y_r is the random variable that counts the number of essential genes in region r and p_i is the probability a gene is essential in state i so that:

If $Z_r = 1$ (non-essential), $Y_r \sim \text{Binomial}(n, p_1)$
 If $Z_r = 2$ (essential), $Y_r \sim \text{Binomial}(n, p_2)$

To reflect prior beliefs about the probability of essential genes in each type of region, Beta priors are placed on p_1 and p_2 which are bounded between 0 and 1. The initial choice of prior of p_1 uses a Beta distribution (3,6) which assumes that non-essential regions are expected to have relatively low proportions of essential genes, while still allowing variability. This generates a right-skewed distribution, as it assumes that as p_1 is for the probability of an essential gene in a non-essential region, there is a much lower likelihood of essential genes being present. This skews the prior to lower probabilities. A plot of this prior is shown on the left side of Figure 4.5. In contrast, a Beta(5,5) distribution is chosen as the prior for p_2 . This choice represents a more balanced belief as it is symmetric around 0.5 which implies that essential genes are relatively common in essential regions. This also reflects greater uncertainty but allows for higher essential gene density in essential regions. A plot of this prior is shown on the right side of Figure 4.5. In both cases, these are weakly informative priors meaning they incorporate some belief about the structure of the data but are not overly restrictive.

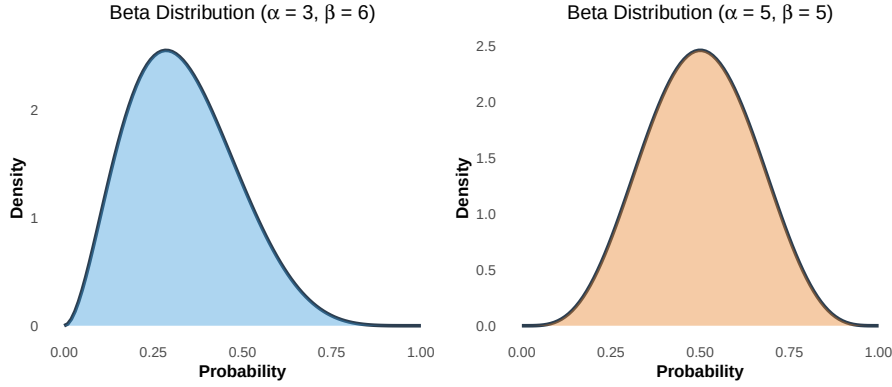


Figure 4.5. Priors for the p_1 and p_2 of the Basic Region HMM. These are beta distributions with more mass placed at lower probabilities for p_1 . p_2 is evenly distributed, reflecting the uncertainty.

The binomial likelihood is the probability of observing a specific number of essential genes in a region, given the assumed essentiality rate for the hidden state. It is given by:

$$P(Y_r = y_r | Z_r = i) = \binom{n}{y_r} p_i^{y_r} (1 - p_i)^{n - y_r}$$

Combining the joint likelihood of the observed data $y = (y_1, \dots, y_{N_r})$ and the hidden state sequence given the model parameters $T = \{\delta, \Gamma, p_1, p_2\}$ gives:

$$P(y, z | T) = \delta_{z_1} \prod_{r=2}^{N_r} \gamma_{z_{r-1}, z_r} \prod_{r=1}^{N_r} P(y_r | z_r, p_{z_r})$$

To perform Bayesian inference, we marginalize over the hidden states to obtain the marginal likelihood of the observed data:

$$P(y|T) = \sum_z P(y, z|T)$$

4.2.3 Model Definitions and Cost Threshold

The nomenclature of the model from Stan is defined in Table 4.3. The previously defined priors for the emission probability encode the belief that $p_2 > p_1$, or that there is a higher probability that the genes in the essential region are often essential. This is compared to genes in non-essential regions that are rarely essential. Initial values are selected with justification for the MCMC sampling, as shown in Table 4.4. The model is fitted using four independent MCMC chains, each with 4,000 iterations. The first 2,000 iterations of each chain are treated as warm-up (or burn-in) to allow the sampler to adapt and stabilize; these iterations are later discarded.

Table 4.3. Parameters and their descriptions for the model

Parameter	Description
tr_from1[1]	Probability of staying in a non-essential region
tr_from1[2]	Probability of moving from non-essential to essential
tr_from2[1]	Probability of moving from essential to non-essential
tr_from2[2]	Probability of staying in an essential region
delta[1]	Probability of starting in a non-essential region
delta[2]	Probability of starting in an essential region
p1	Probability of an essential gene in a non-essential region
p2	Probability of an essential gene in an essential region

Table 4.4. Initial Parameter Values and Justification

Parameter	Value	Justification
p1	0.1	Assume a low probability of an essential gene in a non-essential region
p2	0.9	Assume a high probability of an essential gene in an essential region
tr_from1	c(0.8, 0.2)	Non-essential regions are more common; transitions to essential regions are less frequent
tr_from2	c(0.1, 0.9)	Essential regions tend to stay, and genes move less frequently out of them

Once the model is fit using HMC in Stan, the posterior probabilities of each hidden state (essential or non-essential) are estimated for every genomic region. From the MCMC samples, we obtain an approximation of the posterior distribution $P(z_i = k | y)$ for each state $k \in 1, 2$ and region i , where y denotes

the observed gene essentiality data. These distributions are then summarized by computing the posterior mean probability of each state across the sampled Markov chains.

$$\hat{p}_i(k) = \frac{1}{U} \sum_{u=1}^U p^u(z_i = k|y)$$

where U is the number of posterior samples and $p^u(z_i = k|y)$ is the posterior of state k in the u^{th} MCMC iteration. Focusing on the essential state $k = 2$, these posterior mean probabilities $\hat{p}_i(2)$ are continuous scores reflecting the likelihood that region i is essential. These continuous probabilities can be converted into binary classifications by applying thresholds where region i is classed based on the following:

$$\hat{c}_i = \begin{cases} 1 & \text{if } \hat{p}_i(2) > \text{Threshold} \\ 0 & \text{otherwise} \end{cases}$$

These region-level predictions are then mapped back to individual genes based on their regional assignments. The predictive performance of the model can be determined by comparing the predictions of $\hat{c}_i$ against the true essentiality labels c_i for each gene. By varying the threshold over a continuous range, Receiver Operating Characteristic (ROC) curves can be constructed to evaluate the model's discrimination ability across all possible operating points.

An important consideration is that the consequences of misclassifications could be asymmetric. Misclassifying an essential gene as non-essential (false negative) may incur a higher cost than incorrectly labeling a non-essential gene as essential (false positive). This can be managed by introducing a cost-sensitive decision rule based on Bayesian decision theory, which assumes there is no cost for correct classifications and that there are two costs input into the threshold. These are: a false discovery cost C_{FP} when a non essential gene is classified as essential, and a false non-discovery cost C_{FN} when an essential gene is not classified as such. If p_i is the probability that region i is classed as essential, then classification threshold is given by:

$$p_i > \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{1}{1 + Q}$$

where $Q = \frac{C_{FN}}{C_{FP}}$ is the ratio of the false negative cost to the false positive cost. (Nlebedim, V, et al, 2021). A range of cost ratios are evaluated to give the optimal classification threshold and corresponding metrics for each dataset.

4.2.4 Model Results

The basic HMM is first applied across all replicore datasets to obtain diagnostic and classification results. To determine the optimal region size, the model is fit

at multiple region sizes and evaluated for classification performance and stability. Once the optimal region size is selected, the subsequent sections present posterior estimates based on this configuration.

4.2.5 Effect of Region Size

Region size is a flexible modeling parameter from which the mean posterior probabilities can be extracted. These probabilities may then be used to classify a given region as essential or non-essential depending on an assigned threshold (see Section 4.2.6). Figure 4.6 shows the AUC, entropy, and number of convergence issues as a function of region size for the HMM. Convergence issues are defined as cases where $\hat{R} > 1.01$ (from Eq. (3.2)). As the region size decreases, the AUC generally improves, reflecting the ability of smaller regions to capture local patterns in essentiality with finer resolution. The entropies are calculated from Eq. (3.4), but have variable trend behavior across replichores. For example, the left replichore of *ecoli* has entropies that initially increase at intermediate region sizes before dropping again at the smallest region sizes. In contrast, the left replichore of *saureus* shows the opposite behavior, with slightly higher entropies at smaller region sizes. Convergence is stable across nearly all *ecoli* replichores (except at a region size of 20), whereas the *saureus* replichores show persistent convergence issues, particularly as region size increases. The increasing $\hat{R}$ issues at larger region sizes imply that these are likely over-smoothing the data, mixing essential and non-essential genes in a way that breaks the model's assumptions or makes parameter inference unstable. The results indicate the optimal region size for the model is 5 genes. Lower or higher region sizes generate poorer AUC's values or have convergence issues. A tradeoff for the improved classification performance of the model is the additional computational cost for running at smaller region sizes.

4.2.6 Effect of Threshold Values

The decision threshold can be varied using the Bayesian decision theory framework from Section 4.2.3 and applied to each replichores. Figure 4.7 shows the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR) as a function of the cost ratio Q , where Q represents the relative cost of false negatives to false positives. As Q increases (false negatives become more costly), the decision threshold lowers which increases TPR at the cost of a higher FPR. When Q is small, the model becomes more conservative in assigning positive labels, leading to both lower TPR and FPR as it prioritizes avoiding false positives (Figure 4.7). The best performance for the Basic HMM is observed in the right replichore of *ecoli* (AUC = 0.971) (see Section 3.6.2 for interpretation of AUC), while the left replichore of *saureus* performs worst (AUC = 0.943). Overall, these results indicate that the model has good discriminative ability to classify essential genes though performance is slightly weaker for the *saureus* genome.

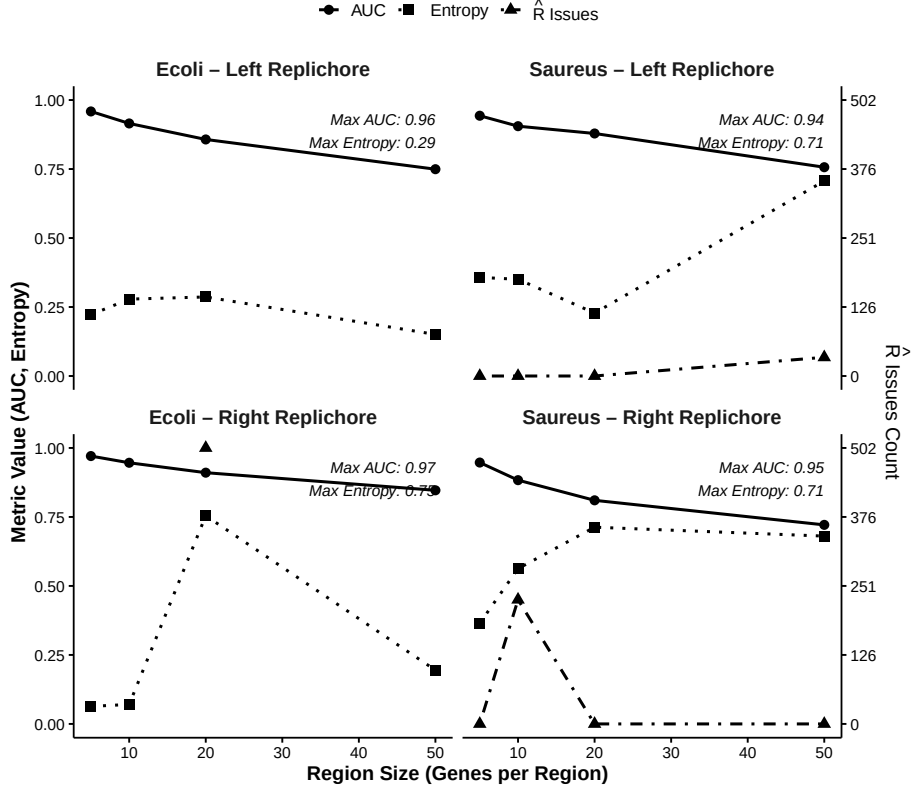


Figure 4.6. Plots of the Basic HMM with region sizes of 5, 10, 20 and 50 across replichores. There is an increase in AUC at region size 5 for all models with the lowest $\hat{R}$ and entropy. $\hat{R}$ is a count of $\hat{R}$ values greater than 1.01.

4.2.7 Model Diagnostics Summary

Table 4.5 presents the parameter diagnostics for the replichores under the Basic HMM (region size = 5). The estimates are stable, as indicated by $\hat{R}$ values below 1.01 and the large effective sample sizes (from Eq. (3.3)). For the left replichore of the *ecoli* genome, the probability of an essential gene occurring in a non-essential region was very low (p_1 , mean = 0.018), while the probability in essential regions was much higher (p_2 , mean = 0.377) (Figure 4.8). Transition probabilities indicate strong persistence in non-essential regions ($\text{tr}_{\text{from } 1}[1] = 0.883$) and moderate persistence in essential regions ($\text{tr}_{\text{from } 2}[2] = 0.391$). Transitions from essential to non-essential regions ($\text{tr}_{\text{from } 2}[1] = 0.608$) were more likely than the reverse ($\text{tr}_{\text{from } 1}[2] = 0.116$), reflecting the class imbalance between essential and non-essential genes and the clustering of essential genes into localized regions. The right replichore showed broadly similar behavior with one exception: essential regions exhibited stronger persistence ($\text{tr}_{\text{from } 2}[2] = 0.558$) compared to the probability of transitioning back to non-essential regions ($\text{tr}_{\text{from } 2}[1] = 0.441$). However, the wide credible intervals around these estimates suggests this difference may not be statistically meaningful.

The *saureus* dataset is similar to *ecoli* in the parameter estimates, but has wider

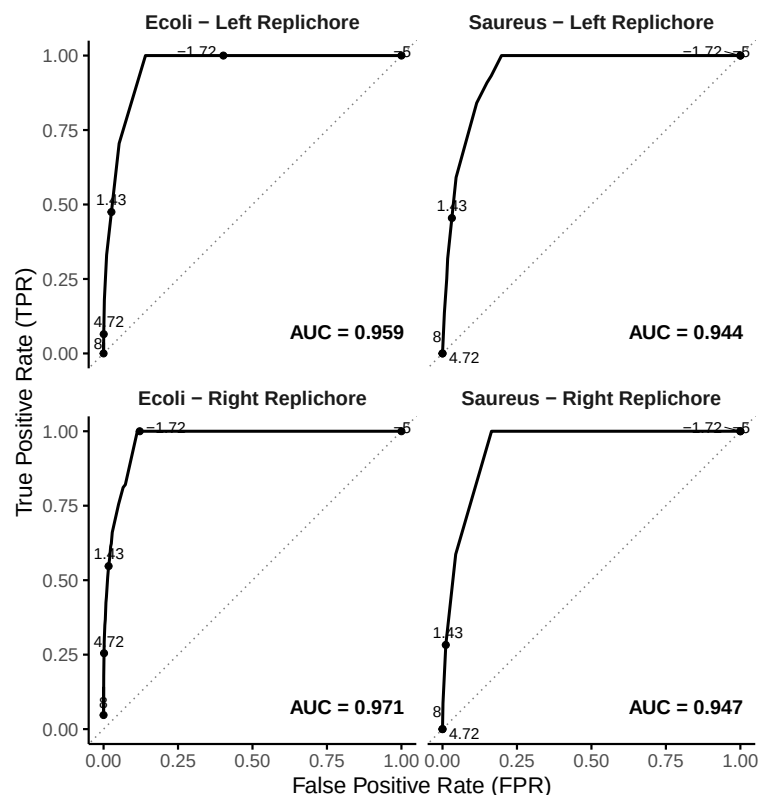


Figure 4.7. ROC curves for the Basic HMM for each replichore, using a region size 5. The cost function is used to show the effect of weighting costs of false negatives to false positives. For ease of reading, all values are in logarithmic base 10 (e.g. 3 on the plot is a value of 1000). The saureus data set has poorer performance based on the AUC compared to ecoli.

posterior uncertainties. This is likely due to the scarcity of data on essential genes, which makes estimates have a wider degree of error than in the ecoli data. The positions of the probabilities are consistent, with p_1 having the lowest probability and the transition probability $\text{tr}_{\text{from } 1}[1]$ having the highest probability. In both cases, this may be attributed to the significant class imbalance in the dataset and the paucity of essential genes.

For all replichores, the model had consistently low probabilities for the essential genes in non-essential regions p_1 and a higher probability in essential ones p_2 . Transition probabilities indicate non-essential regions move to other non-essential regions (due to class imbalance) than essential regions. The parameter estimates were well-supported with high effective sample sizes and convergence diagnostics, though it is anticipated that introducing the covariates insertion density and positional preference to the origin will improve classification.

Table 4.5. $\hat{R}$ and Effective Sample Size for the Basic HMM parameters across datasets and replichores.

Dataset	Replichore	Parameter	Effective Sample Size	Rhat
Ecoli	Left	p1	9411.378	1.000
Ecoli	Left	p2	9944.131	1.000
Ecoli	Left	tr_from1[1]	9720.190	1.000
Ecoli	Left	tr_from1[2]	9720.190	1.000
Ecoli	Left	tr_from2[1]	15936.592	1.000
Ecoli	Left	tr_from2[2]	15936.592	1.000
Ecoli	Right	p1	14483.664	1.000
Ecoli	Right	p2	13393.376	1.000
Ecoli	Right	tr_from1[1]	15370.523	1.000
Ecoli	Right	tr_from1[2]	15370.523	1.000
Ecoli	Right	tr_from2[1]	16719.778	1.000
Ecoli	Right	tr_from2[2]	16719.778	1.000
Saureus	Left	p1	10700.309	1.000
Saureus	Left	p2	9601.983	1.000
Saureus	Left	tr_from1[1]	14092.814	1.000
Saureus	Left	tr_from1[2]	14092.814	1.000
Saureus	Left	tr_from2[1]	10800.969	1.000
Saureus	Left	tr_from2[2]	10800.969	1.000
Saureus	Right	p1	7955.401	1.001
Saureus	Right	p2	7216.064	1.001
Saureus	Right	tr_from1[1]	9658.674	1.001
Saureus	Right	tr_from1[2]	9658.674	1.001
Saureus	Right	tr_from2[1]	8466.178	1.001
Saureus	Right	tr_from2[2]	8466.178	1.001

4.2.8 Further Diagnostics

The convergence and stability of the posterior sampling for the Basic HMM can also be examined through traceplots for the key model parameters. Figures 4.9-4.11 show the trace plots for the emission probabilities p_1 and p_2 and the transition probabilities for $\text{tr}_{\text{from } 1}[1]$, $\text{tr}_{\text{from } 2}[1]$, $\text{tr}_{\text{from } 1}[2]$, and $\text{tr}_{\text{from } 2}[2]$. Across all replichores, the chains show stable mixing behaviour and explore the parameter space without any flat segments or visible trending. The chains from different initializations overlap consistently which indicates that the MCMC sampling achieved good convergence for the emission parameters. p_1 has tighter variation compared to p_2 , which is consistent with the narrower credible intervals for p_1 . These plots indicate that the posterior estimates are generated from a reliable base and are not affected by convergence issues.

The parameter autocorrelation can be assessed by the effective sample size and autocorrelation function (ACF) plots. Given the large number of parameters, multiple chains and replichores, the ACF is assessed using the Lag-1 ACF which is a measure of how strongly each sample in a Markov chain is correlated with the previous sample. A low lag-1 ACF indicates that consecutive draws are largely independent, which is desirable for efficient exploration of the posterior.

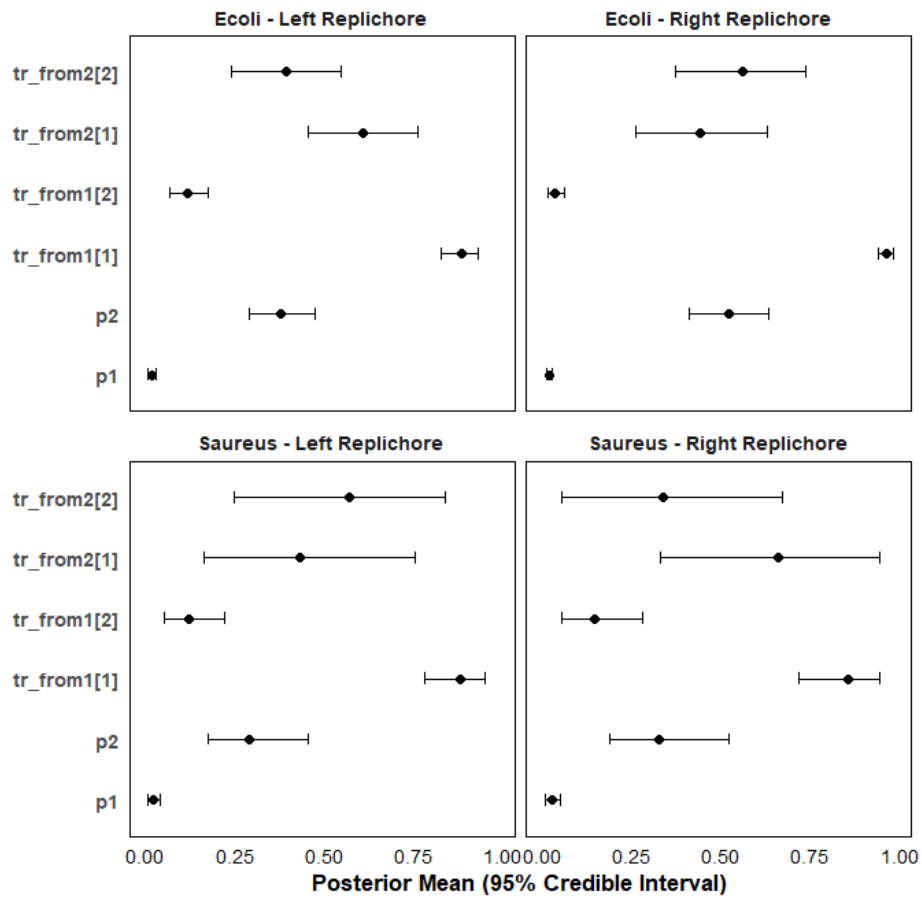


Figure 4.8. Estimates of the posterior means for each replicore, with 95% confidence intervals (using Basic HMM with a region size of 5 genes). There is a relatively precise estimate of p1 compared to the wide confidence intervals for the transition probabilities in the saureus dataset.

This is shown in Figure 4.12, which has relatively low values for the Lag-1 ACF values. Coupled with the high effective sample size noted in Table 4.5, this indicates low levels of autocorrelation meaning the MCMC chains are mixing well and providing nearly independent samples for parameters. Full ACF plots for key parameters are available in Chapter 6.

4.3 Hidden Markov Models (Additional Covariates)

The Basic HMM can be extended by incorporating the covariates insertion density (InsDens) and distance from the replication origin (DFRO) into the model. Each HMM assumes a sequence of hidden essentiality states and observed data generated from state-specific emission distributions.

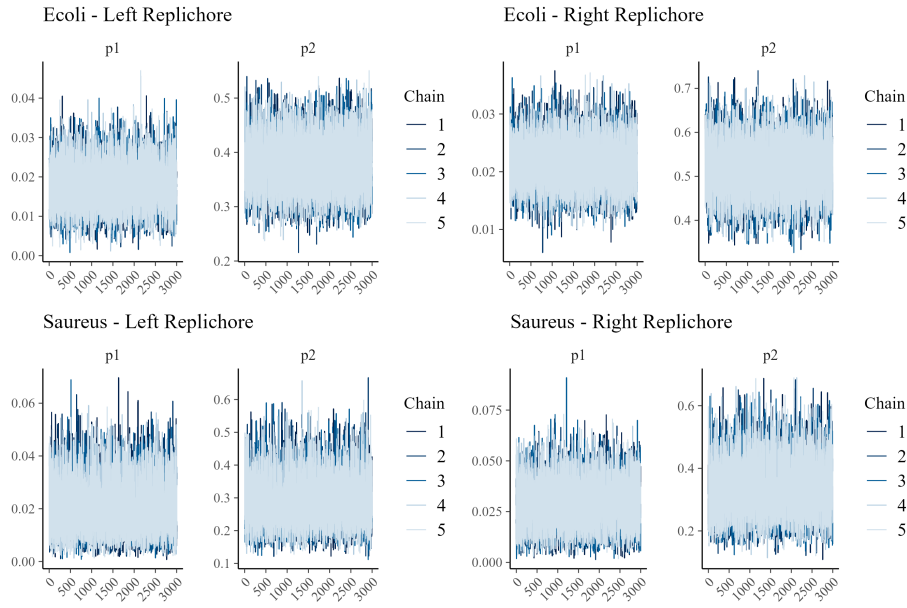


Figure 4.9. Trace plots of the parameters p1 and p2 for each of the replichores using the Basic HMM (region size 5). The plots show good mixing of the chains with no directional drift or repeated values.

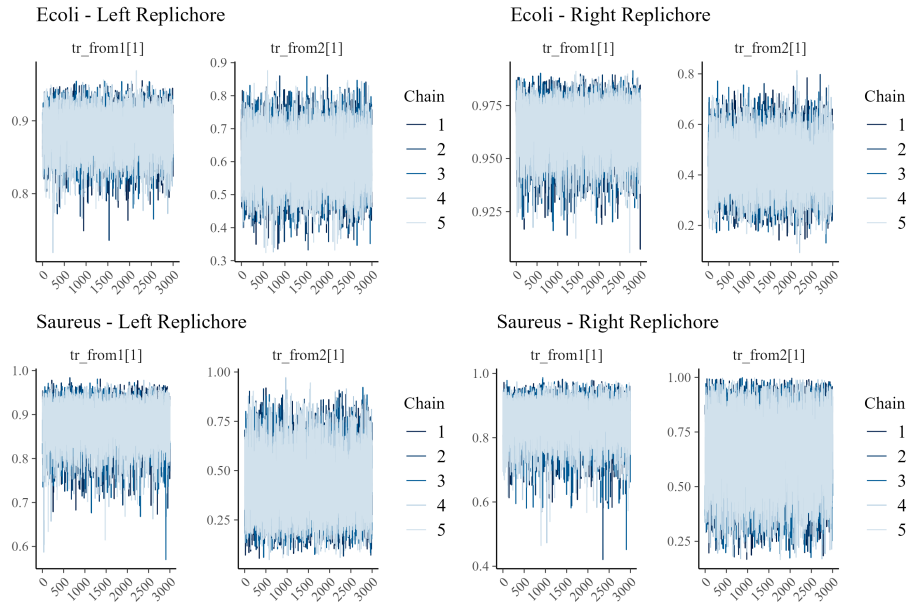


Figure 4.10. Trace plots of the parameters trfrom1(1) and trfrom2(1) for each of the replichores using the Basic HMM (region size 5). The plots show good mixing of the chains with no directional drift or repeated values.

4.4 Insertion Density

The insertion density is calculated for each gene as the number of unique transposon insertion sites normalized by the gene length, which reflects the relative

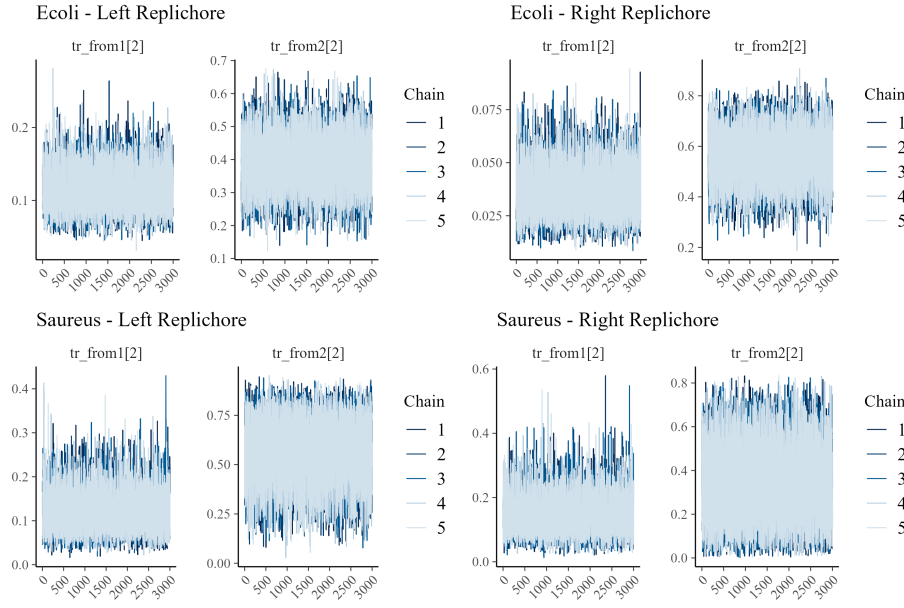


Figure 4.11. Trace plots of the parameters $\text{trfrom1}(2)$ and $\text{trfrom2}(2)$ for each of the replichores using the Basic HMM (region size 5). The plots show good mixing of the chains with no directional drift or repeated values.

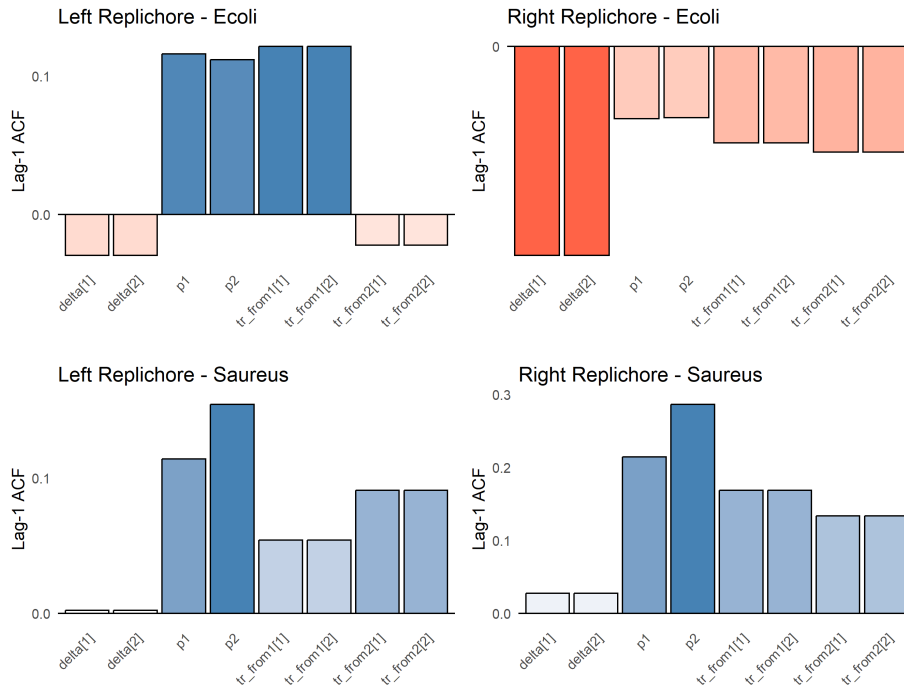


Figure 4.12. Lag-1 ACF Plots for some key parameters of the Basic Region HMM, split by replichire. The values are low which indicates low autocorrelation.

tolerance of gene size to insertions (Eq. (2.1)). Genes with low insertion densities are more likely to be essential, as essential genes tend to tolerate fewer

insertions. The distribution of the insertion densities for each genome and replichore are plotted in Figures 4.13 and 4.14. These are split by the essentiality of the genes. For each group and genome region, multiple candidate probability distributions are fitted to determine the emission probabilities in the HMM. The fitting process involves estimating distribution parameters using maximum likelihood and evaluating the goodness of fit. Different fitted distributions overlaying the histograms are also shown in Figures 4.13 and 4.14.

The proposed distributions are the Log-Normal, Gamma, and Beta distributions. Exponential distributions are also fitted for the essential gene insertion densities, as previous modeling efforts with the *ecoli* bacterium have adopted the exponential and gamma distributions for the priors (Larivière, D, et al, 2021). The insertion density is bounded between 0 and 1, with no values at exactly 0 or 1. The insertion densities for the essential genes are clustered at very low values, indicated by the right skewed plot in all of the replichores. Additionally, there are long tails caused by some essential genes at higher insertion densities (potential spurious results from TraDIS). These features support the use of the selected distributions for the emission priors, particularly the Beta distribution. The data exhibits multimodal distributions and wide variance, which complicate fitting smooth parametric models. This complexity is shown in the *saureus* dataset, where the scarcity of essential genes and multiple peaks in the histograms create fitting difficulties. As histograms depend on arbitrary binning choices, the visual distribution shape can vary which will also impact the fitting results.

The plotting shows that each proposed distribution fits the data generally well for the *ecoli* dataset. The essential genes have a strong right skew which is modeled well by all distributions, while there are poorer fits for the non-essential genes. The *saureus* dataset does not fit well to the distributions, particularly for the non-essential genes. The Beta distribution was chosen to model insertion density as an emission feature in the HMM, due to the data being bounded between 0 and 1 and its flexibility in capturing skewed distributions. To allow for regional variation, the shape parameters of the Beta distribution are assigned hierarchical priors. The region-level Beta parameters are derived from a hierarchical structure where the population-level precision parameters are governed by hyperpriors (Section 4.2.1). This approach has been explored in previous literature, with a specific case using Gamma hyperpriors on the Beta distributions (Nlebedim, V, et al, 2021).

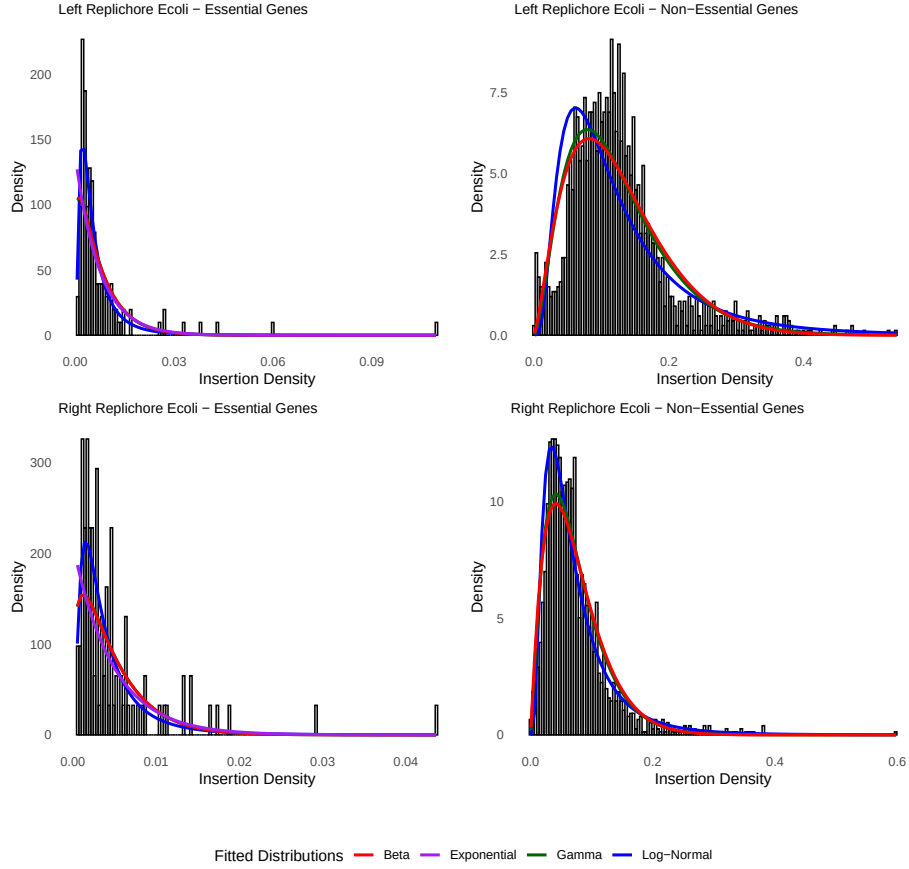


Figure 4.13. Histograms of the insertion density distribution for the ecoli replichores, split by essential status. Multiple distributions are fitted, of which the Beta distribution adapts well to the skew and peaks of the data.

4.4.1 Model Development

The Basic HMM defined to classify genomic regions based on observed essentiality can be extended with the addition of insertion density. For this model, the genomic regions have two data types $y_r^{(1)}$ and $y_r^{(2)}$. $y_r^{(1)}$ is the essentiality of regions, modeled as binomial counts in the previous basic HMM. $y_r^{(2)}$ is the insertion density, modeled as a Beta distribution.

Let $S_r \in \{1, 2\}$ denote the hidden state of a region $r = 1, \dots, N_r$, where $S_r = 1$ is a non-essential region and $S_r = 2$ is an essential region.

Transition Probabilities and Initial State Distribution

These are the same as described in Section 4.2.2.

Observed Data Likelihoods

The observed data consists of two parts: binomial essentiality counts and insertion density.

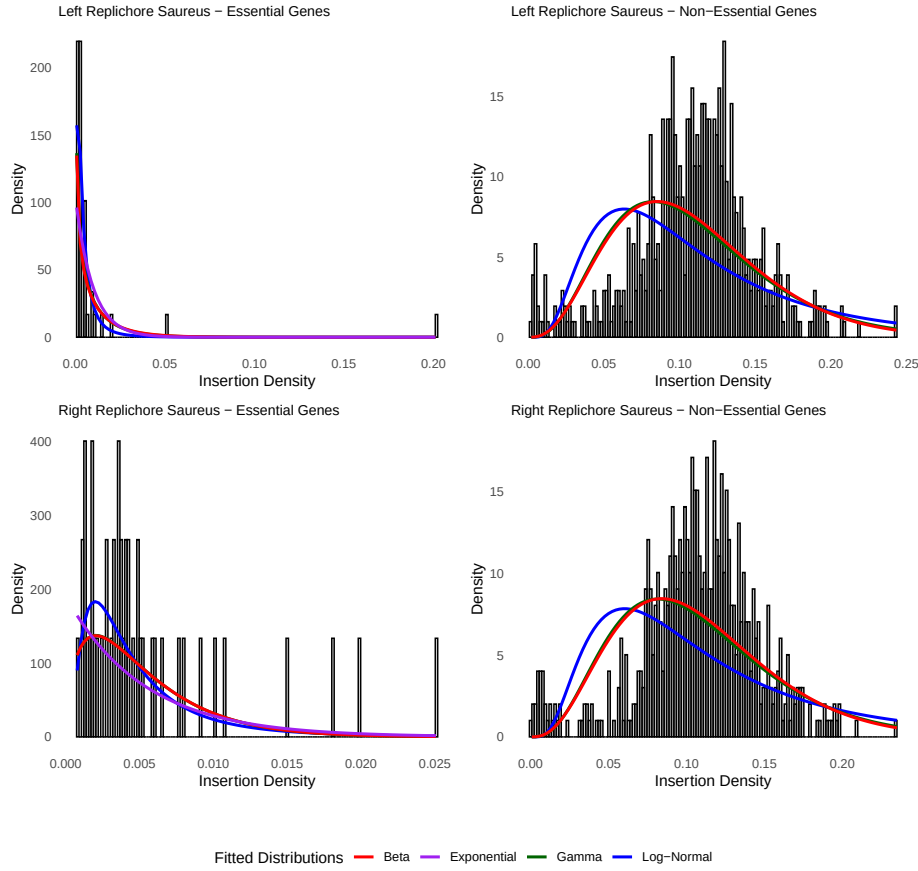


Figure 4.14. Histograms of the insertion density distribution for the saureus replichores, split by essential status. Multiple distributions are fitted though there is difficulty due to the low number of essential genes and the small peaks on the non essential genes. The beta distribution appears to manage the shape of the data best

Binomial Essential Counts

Let:

- $y_r^{(1)} = \text{numess}_r$: number of essential genes in region r .
- $N_r = \text{numtrials}_r$: total number of trials (total number of insertions in the region) and then:

$$y_r^{(1)} | S_r = k \sim \text{Binomial}(N_r, p_k)$$

where:

- $p_k \in (0, 1)$: probability of observing an essential insertion in state k .

- Prior on p_k : $p_k \sim \text{Beta}(2, 2)$
- Constraint: $p_1 < p_2$. The constraint reflects the assumption that essential regions (state 2) have a higher probability of observing essential insertions.

Insertion Density (Beta Likelihood)

Let $y_r^{(2)} \in (0, 1)$ be the insertion density for region r . This can be modelled as a Beta distribution:

$$y_r^{(2)} | S_r = k \sim \text{Beta}(\alpha_{k,r}, \beta_{k,r})$$

Instead of directly defining the Beta shape parameters, we define them as:

- Mean: $\mu_{k,r} \in (0, 1)$
- Precision: $\phi_{k,r} > 0$

By defining the Beta distribution using its mean and precision, the central tendency (insertion rate) can be separated from the variability (region-level noise). This makes the model easier to interpret and allows for uncertainty modeling. These relate to the shape parameters by:

$$\alpha_{k,r} = \mu_{k,r} \phi_{k,r}$$

$$\beta_{k,r} = (1 - \mu_{k,r}) \phi_{k,r}$$

This allows separate control of the mean and variance of the Beta distribution for each region.

Random Effects for Regional-Level Variation

We allow $\mu_{k,r} \in (0, 1)$ and $\phi_{k,r} > 0$ to vary across regions within each state k . From the essential gene perspective, $\mu_{k,r}$ can be interpreted as the probability of observing an insertion in region r given hidden state k . For essential states, $\mu_{k,r}$ will be close to zero, while for non-essential states, it will be higher.

$$\mu_{k,r} = \text{logit}^{-1}(\text{logit}(\mu_k) + \sigma_{\mu,k} z_{\mu,k,r}), \quad z_{\mu,k,r} \sim N(0, 1)$$

where: μ_k = global mean for state k , $\sigma_{\mu,k}$ = standard deviation for the random effects, $z_{\mu,k,r} \sim N(0, 1)$ = standard normal noise per region. A logit function is used because it transforms the mean parameter $\mu_{k,r}$, which is constrained to the interval (0,1) onto the real line. This allows modeling the regional variation with additive Gaussian noise $z_{\mu,k,r} \sim N(0, 1)$ in an unconstrained space. After adding this noise, and inverse logit function is applied to map the value back

to $(0,1)$, ensuring that $\mu_{k,r}$ remains a valid probability. The precision on a log scale is given by:

$$\phi_{k,r} = \exp(\log(\phi_k) + \sigma_{\phi_k} z_{\phi,k,r})$$

where:

- ϕ_k : is the global precision for state k .
- $\sigma_{\phi,k}$: standard deviation for precision random effects.
- $z_{\phi,k,r} \sim N(0, 1)$: standard normal noise.

The random effects $z_{\mu,k,r}$ and $z_{\phi,k,r}$ allow for regional variability in insertion behavior, even within the same hidden state. This acknowledges that biological and technical variation may cause differences in observed densities that cannot be explained by the state alone.

Population Parameters

For each hidden state $k \in \{1, 2\}$, we place the following priors:

- $\mu_k \sim \text{Beta}(2,2)$: global mean insertion density. The Beta(2,2) prior is symmetric and relatively uninformative but slightly favors values around 0.5. This choice reflects reasonable uncertainty about the global mean insertion density in each state, allowing the data to guide inference while preventing extreme estimates.
- $\phi_k \sim \text{Gamma}(2,0.1)$: global precision. The precision parameter ϕ_k is strictly positive and controls the concentration of the Beta distribution. The Gamma distribution is a standard conjugate prior for positive scale parameters. Choosing shape = 2 and rate = 0.1 is a moderately informative prior that favors moderate precision values but allows enough flexibility to adapt to the data.
- $\sigma_{\mu,k} \sim \text{Half Normal}(1)$: standard deviation of random effects. The half-normal prior constrains these standard deviation hyperparameters to be positive, reflecting that variability can't be negative. The scale of 1 is chosen to be weakly informative, allowing for a reasonable range of variability in regional effects without being overly permissive.
- $\sigma_{\phi,k} \sim \text{Half Normal}(1)$: standard deviation of precision random effects. See above on standard deviation on random effects.

The hyper parameters $\sigma_{\mu,k}$ and $\sigma_{\phi,k}$ control the amount of regional variation around the state-level mean and precision, respectively. If the data has little variation, the posterior will shrink these toward zero via the Half-Normal prior.

This is hierarchical Bayes model, where:

- State-level parameters (e.g. μ_k , ϕ_k) govern average behavior in each class.

- Region-specific values $(\mu_{k,r}, \phi_{k,r})$ vary randomly around those global parameters.
- Observed data $y_r^{(1)}, y_r^{(2)}$ are modeled conditionally on the hidden state S_r and the region-level parameters.

4.4.2 Prior Distribution Comparisons

The results for the different parameterisations are presented in Table 4.6. For each model and replichore, the AUC is reported for predictive performance, along with the number of parameters exceeding the $\hat{R}$ convergence threshold as an indicator of model stability. The basic HMM is provided in the table as a benchmark to compare to the extended models. The models with no convergence issues are the Basic HMM, the Log-Normal model and the Exponential-Gamma model. These latter two models have lower AUCs even though they have no convergence issues, suggesting that the priors of the model are not adequately capturing the complexity or variability of the insertion density.

The Beta and Hierarchical Beta models achieve similar AUC values to the Basic HMM. Additionally, the Beta model has convergence issues as indicated by the large number of parameters with $\hat{R}$ values well above the acceptable threshold. This instability suggests incorrect prior and parameter selection or insufficient information to reliably estimate the parameters. The ROC curves are shown for the Hierarchical Beta model in Figure 4.15. The model demonstrates high AUC's but has difficulty with convergence on the right replichore of the *ecoli* dataset. The incorporation of the insertion density covariate as an emission probability in the model does not substantially improve the model using this parameterisation, with lower or similar values to the basic HMM. This suggests the covariate is adding more noise to the model than signal, and that the magnitude of the values are so low that they do not contribute to improved performance.

4.5 Distance from the Replication Origin

The extended HMM using the distance from the replication origin (DFRO) builds directly on the structure of the basic model, which models the genome as a sequence of regions, each belonging to one of two hidden states. The spatial variation in essentiality along the replichores is captured with the scaled distance from the replication origin $d_n \in (0, 1)$ given by:

$$d_n = \min \left(\max \left(\frac{|s_n - s_o|}{s_{\max}}, \epsilon, 1 - \epsilon \right) \right)$$

where $s_o = 0$ and $\epsilon = 10^{-5}$. Here, s_0 is the position of the replication origin, and ϵ is a small constant introduced to prevent scaled distances from taking on

Table 4.6. Insertion Density Model Performance for each prior. The Rhat counts correspond to the number of parameters with convergence issues for each model (values > 1.01). Changing the priors can lead to different results, suggesting there may not be sufficiently informative to override prior influence.

Model	Ecoli				Saureus			
	Left Replichore		Right Replichore		Left Replichore		Right Replichore	
	Rhat	AUC	Rhat	AUC	Rhat	AUC	Rhat	AUC
HMM Basic	0	0.959	0	0.971	0	0.943	0	0.947
Log-Normal	0	0.808	0	0.802	0	0.758	0	0.853
Beta	1991	0.958	2020	0.969	0	0.947	701	0.613
Hierarchical	0	0.955	344	0.969	2	0.936	0	0.944
Bayes with Beta								
Exponential-Gamma	0	0.530	0	0.971	0	0.844	0	0.700

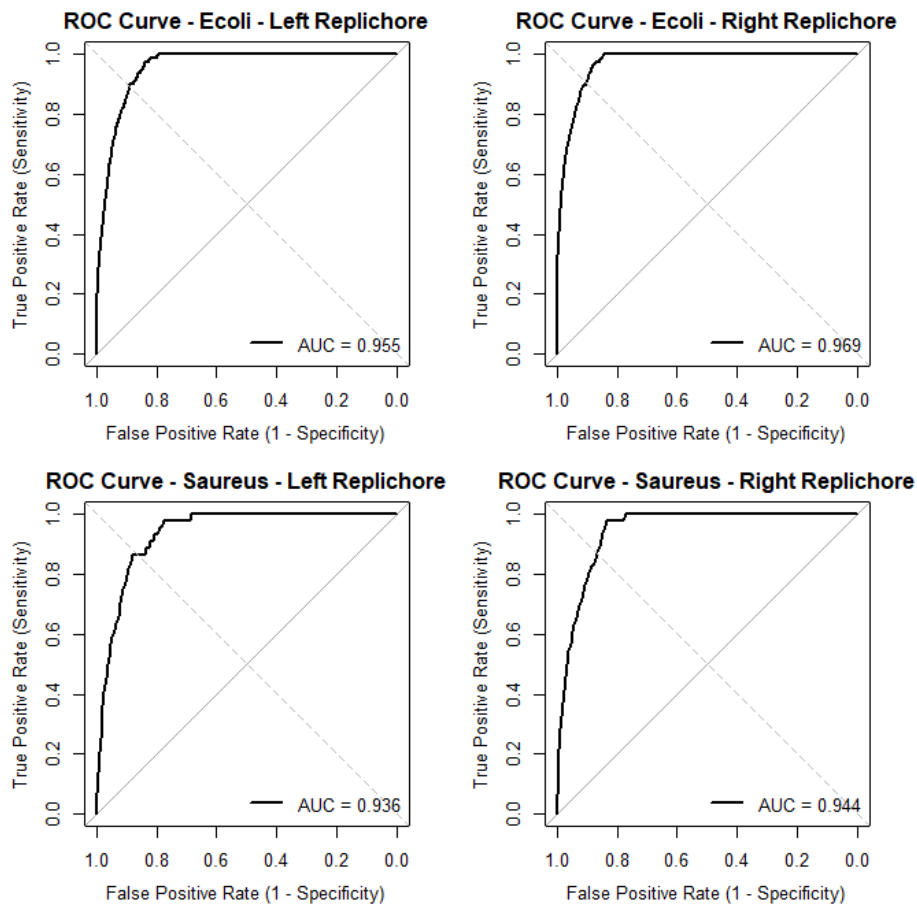


Figure 4.15. ROC Curves of the Hierarchical Beta model for each of the four replichores. We note a poorer performance on AUC for the left replichore of the saureus dataset.

boundary values 0 or 1. This is required for prior stability. For the prior specification, the empirical distribution of scaled distances was examined separately for each replichore and gene essentiality class. Standard probability distributions were fitted to the data and visualised as density lines (Figures 4.16-4.17). The histogram plots for the *ecoli* replichores show varied behavior for the scaled distances, with the essential genes showing multiple potential modes and the non-essential genes having almost uniform behavior with no clear peak. The *saureus* data has even more fluctuating behaviour with multiple modes for the essential genes and erratic distributions for the non essential genes. The most flexible distribution for the data is a Beta distribution, which models the non-essential genes well. The parameters for the non-essential genes are almost 1, reflecting an almost uniform distribution of the data.

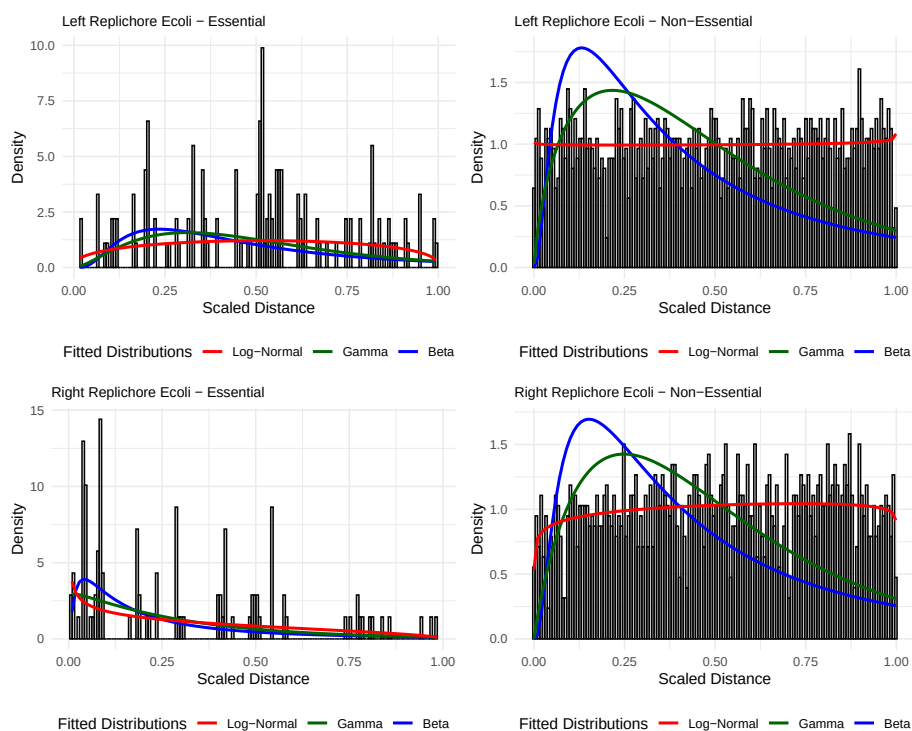


Figure 4.16. Histogram plots of the scaled distance for each *ecoli* replichore, split by essential status. Multiple distributions are fitted to the data.

4.5.1 Model Development - Distance from the Replication Origin

The extension of the basic HMM to incorporate distance from the replication origin with region-specific variation in the emission probabilities by allowing the success probability itself to vary as a function of the scaled distance from the replication origin. This differs with the insertion density model, where the Beta

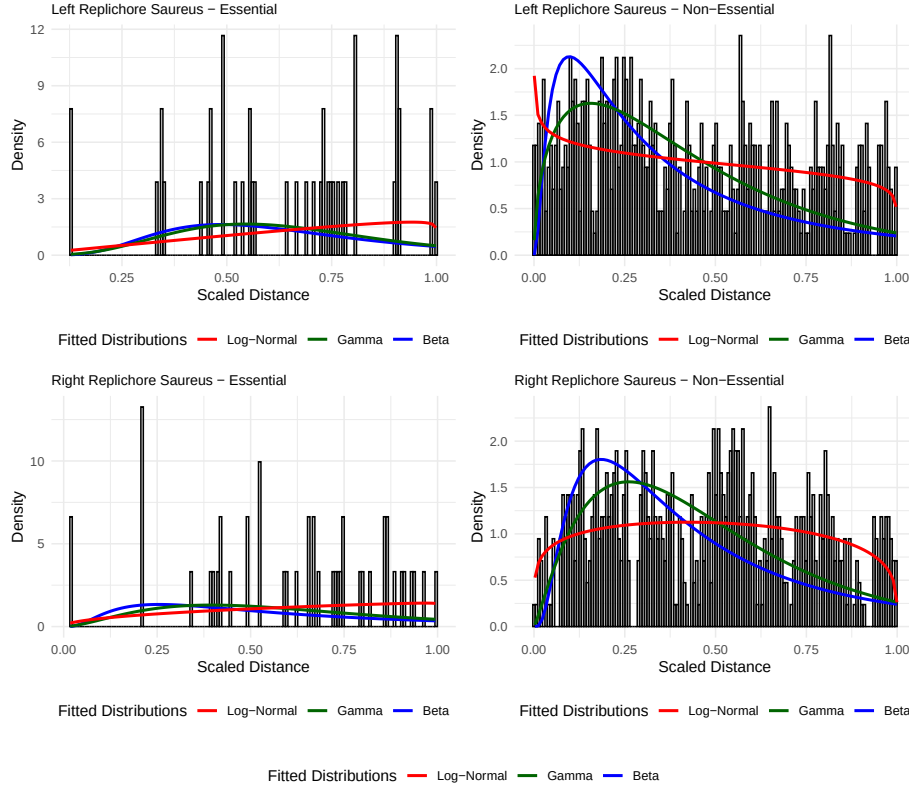


Figure 4.17. Histograms of the scaled distance for each saureus replichore, split by essential status. Multiple distributions are fitted to the data. The beta distribution has a better fit to the shape of the data than the other distributions.

distribution is used to model a separate continuous signal rather than overdispersion of a single Binomial process. Attempts with a similar construction to the insertion density model results in poor convergence for all replichores and was discarded. The Beta-Binomial framework is appropriate for modeling essential gene counts with overdispersion linked to distance, whereas insertion density is modeled as a separate Beta-distributed variable. The model construction is detailed below.

Observed Data

For each genomic region $r = 1, \dots, N_r$, we have y_r = number of essential genes, $d_r \in (0, 1)$ = scaled distance from the replication origin for each region and the hidden state for region r is denoted by $S_r \in \{1, 2\}$.

Transition Probabilities and Initial State Distribution

These are the same as described in Section 4.2.2.

Emission Model

For each state $S_r \in \{1, 2\}$ at region r , the emission distribution for the observed

essential gene counts y_r (out of M_r total genes per region) conditional on state $S_r = s$ is a Beta-Binomial:

$$Y_r | S_r = s \sim \text{BetaBinomial}(M_r, \alpha_{s,r}, \beta_{s,r})$$

The parameters $\alpha_{s,r}$ and $\beta_{s,r}$ are functions of region-level covariates and are defined through a reparameterization using the mean and precision of the Beta distribution. These are given by:

- The baseline success probability $\mu_s \in (0, 1)$
- The baseline precision $\phi_s > 0$
- The covariate effects $\gamma_s, \kappa_s \in \mathbb{R}$
- The covariate d_r

Parameters of the Beta Distribution

The mean probability of essentiality for state S at region r is modeled on a logit scale:

$$\eta_{s,r} = \text{logit}(\mu_s) + \gamma_s(d_r - 0.5)$$

$$\mu_{s,r} = \frac{1}{1 + e^{-\eta_{s,r}}}$$

Here, $\mu_s \in (0, 1)$ is the baseline essentiality probability for state s , while $\gamma_s \in \mathbb{R}$ is the effect of distance from the replication origin on this mean. The logit link is used to ensure probabilities remain within the valid $(0, 1)$ range while permitting a linear dependence on distance in the transformed space. Centering distance at 0.5 makes μ_s interpretable as the essentiality probability at the genome midpoint, and reduces correlation between the baseline and slope parameters for stable estimation.

The precision $\phi_{s,r}$ is also allowed to vary with genomic distance:

$$\log(\phi_{s,r}) = \log(\phi_s) + \kappa_s d_r$$

Here, $\phi_{s,r} > 0$ is the baseline precision, and $\kappa_s \in \mathbb{R}$ is the effect of distance on precision, which allows the model to accommodate uncertainty about essentiality rates across the genome. Then the Beta shape parameters are computed as:

$$\alpha_{s,r} = \mu_{s,r} \phi_{s,r}$$

$$\beta_{s,r} = (1 - \mu_{s,r}) \phi_{s,r}$$

Prior Distributions

$\mu \sim \text{Beta}(2, 2)$: Baseline Mean

$\log \phi_{s,r} \sim \text{Normal}(1, 1)$: Baseline Precision

$\gamma_s \sim \text{Normal}(0, 3)$ and $\kappa_s \sim \text{Normal}(0, 1)$: Covariate Effects

The performance of the model on the replichores is shown in Figure 4.18. There is no improvement in the AUC compared to the basic HMM with significant convergence issues for the left replichore of the *ecoli* genome. Given the inconsistent and multimodal distributions of these parameters and the weak association between the covariate and essentiality, distance from the replication origin was not incorporated further into the HMM.

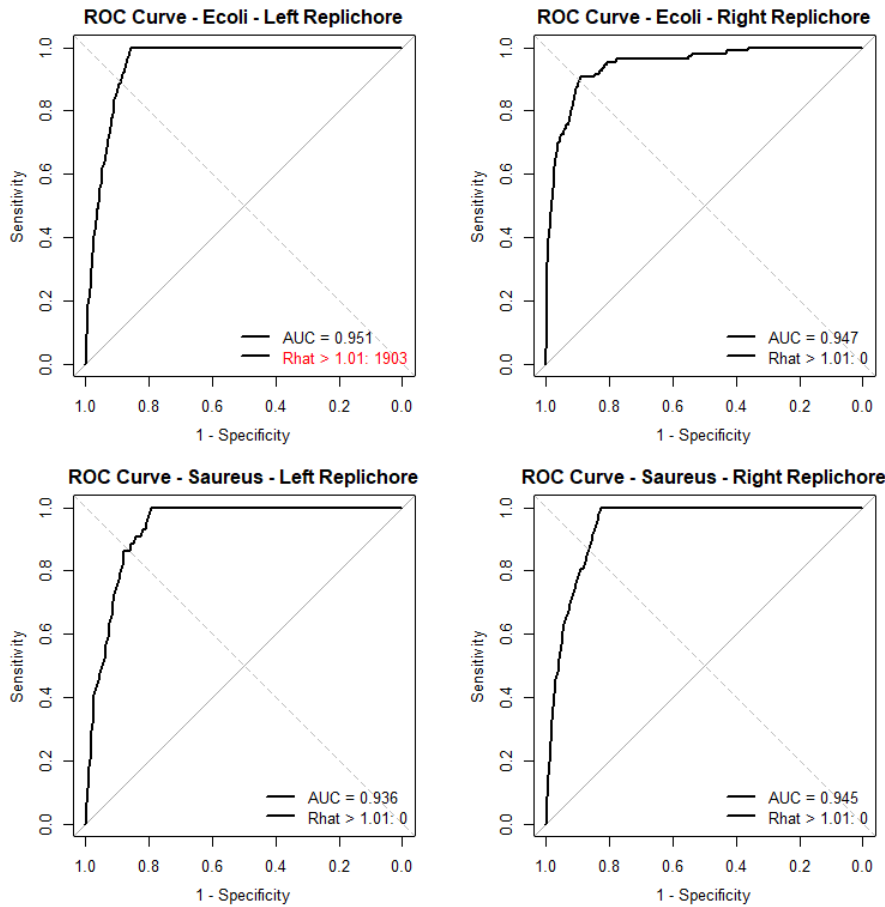


Figure 4.18. ROC curves for each of the replichores using the DFRO only model. We note poor predictions for the left replichore of *ecoli* using the model.

4.6 Data Simulation

The performance of the HMMs can be assessed using synthetic data. Due to the structured and spatially dependent nature of genomic data, standard cross-

validation (e.g. leaving out individual genes or regions) is not appropriate, as random splits disrupt dependencies and produce misleading performance estimates. Initial assessment of the model performed in-sample evaluation by applying the model to different replichores, which serves as a biologically meaningful check of generalizability. Standard cross-validation, reported in Chapter 6), yield AUCs barely above random guessing (0.5).

Because the data cannot be randomly shuffled or rearranged without losing key features, such as the degree of clustering, the synthetic data is generated to retain these properties while allowing controlled variation. Biological data contains many distinct patterns arising from bacterial growth, some specific to the species or environment. The following approach will not capture every property of the data as some are hidden, while some features that are present in a bacterium (such as GC content) are not available in the dataset.

The functions used to generate the simulated data are detailed in Chapter 6. The following selected features of the simulation are allowed to vary (examples shown in Figure 4.19):

- **Gene count and proportion of essential genes:** The simulated data retains the gene count of the original dataset, which can be scaled by a gene count multiplier m_g . Let N denote the total number of genes in the simulation, and then $N = \text{round}(N_{obs} \cdot m_g)$, where N_{obs} is the number of observed genes. For example, 2000 genes with 5% essential genes can be scaled to 4000 genes with the same proportion. Alternatively, the proportion of essential genes P_{ess} can be taken as default from the replichore or adjusted by a proportion multiplier m_p . This generates $N_{ess} = \text{round}(N \cdot P_{ess})$ essential genes with the remainder non-essential genes $N_{noness} = N - N_{ess}$. For example, the class imbalance can be skewed further so that fewer essential genes are present in the simulated dataset.
- **Clustering structure of essential genes:** Clusters of essential genes are generated using the empirical run-length distribution of consecutive essential genes in the real dataset:

$$p_{run}(r) = \frac{\text{count of run lengths } r}{\sum_r \text{count of run lengths } r}$$

Each run length R is scaled by a random factor $U(1, c_{cluster})$, where $c_{cluster}$ is the cluster size factor. Non-essential runs are inserted between essential clusters with lengths drawn from a shifted geometric distribution $R_{noness} \sim 1 + \text{Geom}(p_g)$. This feature enables the data to retain a similar local spatial dependency among essential genes as the original data. The cluster size factor can be used to change the size of clusters, making the run lengths longer or shorter as required.

- **Distance from Replication Origin:** A key spatial feature in bacterial genomes is the distance of each gene from the origin of replication. Start positions for essential and non-essential genes are sampled independently

from the empirical cumulative distribution functions (CDFs) of their observed positions:

$$X_{ess} \sim F_{start,ess}^{-1}(U(0,1)), X_{noness} \sim F_{start,noness}^{-1}(U(0,1))$$

The simulation supports three modes of spatial bias: “uniform”: no status-specific positional bias, “essential_near_origin”: essential genes are more concentrated near the origin and “nonessential_near_origin”: non-essential genes are nearer the origin. The mean and spread of distances can be controlled with a toggeable factor, including a `distance_spread_factor`, which controls the positional skew. For example, the simulated data can use the default positions of the genes in the replichores, or shift the essential genes closer or further away from the origin (like in the *saureus* dataset).

- **Insertion Density Assignment:** Insertion densities for essential and non-essential genes are sampled separately from their empirical distributions I_{obs} in the real data and rescaled so that the mean and standard deviation are multiplied by user-specified factors m_{ess} and s_{ess} (and the same for non-essential). This rescaling is performed by standardising the sampled values, applying the new standard deviation and shifting the new mean. For example, the distributions of insertion densities can be moved closer together, weakening the signal from the feature, or placed further away allowing stronger differentiation between essential and non-essential genes.

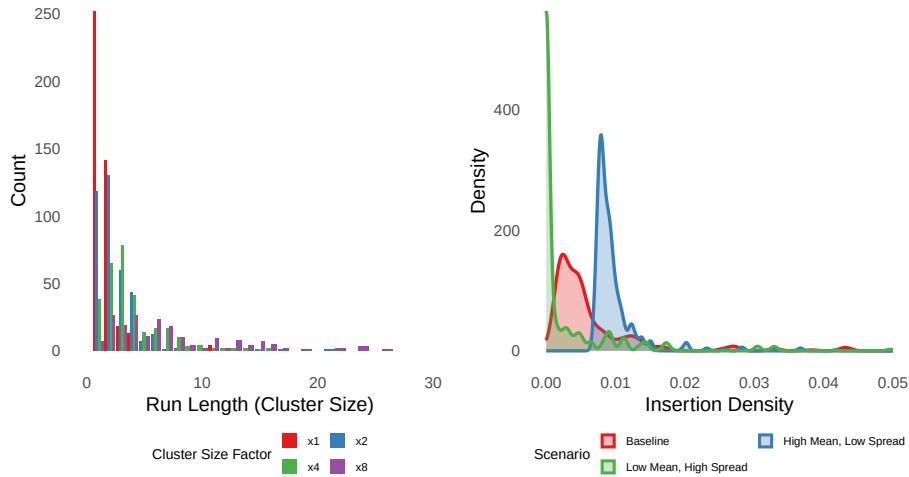


Figure 4.19. Simulated data example using the left replichore of *ecoli*. The left plot is the run lengths scaled by cluster size factors leading to longer runs of essential genes. The right plot is the insertion density of essential genes by changing the mean and spread. This is truncated to 0.05 for visibility.

4.6.1 Basic Region HMM Simulation

These functions can be applied to the Basic Region HMM to simulate data with different genomic features. The degree of clustering is altered by scaling the run length and the proportion of essential genes. Figures 4.20 and 4.21 show the ROC curves for each of the replichores. When the simulated data are generated with the same run length distribution as the observed data, there is a sizeable drop in AUC across all replichores. This suggests that the predictive performance of the model on the real data is not necessarily tied to the prevalence of essential genes, but instead the non-random spatial clustering of essential genes along the genome. This is supported when varying the essential gene proportion, where the AUC's consistently rise as the proportion of essential genes approaches 1%. There may be some improvement caused by the increasing prevalence of non-essential genes, where even a negative guess results in a higher accuracy. As the run length increases, there is a clear positive correlation with the AUC for the same proportion of essential genes, indicating that as the clustering becomes more pronounced the model can more accurately predict the positive class. This is expected using a HMM compared to a typical classifier used on independent data (such as logistic regression), where this feature is exploited by the model. There are isolated convergence issues when varying the run length, particularly for the saureus data due to the smaller genome size leading to shorter effective runs and reduced information for stable parameter estimation.

Despite these convergence problems, the overall pattern remains consistent across replichores with model performance strongly shaped by the spatial structure of essential genes. For both simulated datasets, the AUC is generally high indicating good discrimination of essential regions. As the feature is more strongly expressed, the classification performance increases correspondingly.

4.6.2 Region HMM with Insertion Density Simulation

The basic region model does not incorporate insertion density as a feature. As a result, varying insertion density in the simulated datasets is not expected to produce any changes in model performance beyond random variation. The contribution of insertion density to classification is investigated by applying the InsDens HMM to three simulated datasets, with parameter settings outlined in Table 4.7. The corresponding ROC curves for both the simulated and observed datasets are presented in Figure 4.22. The model shows reduced performance on the simulated datasets relative to the observed data, though there are exceptions such as the first two simulations on the left replichore of the saureus genome.

The worst performance is from the third simulation, where the distributions of the essential and non-essential gene insertion densities are brought closer together. This is anticipated given it will lead to more overlapping points, creating a weaker signal for the model to exploit. Overall, the model retains high AUC's but has some notable convergence issues and erratic plots, particularly for the saureus genome. This suggests the paucity of essential genes is negatively

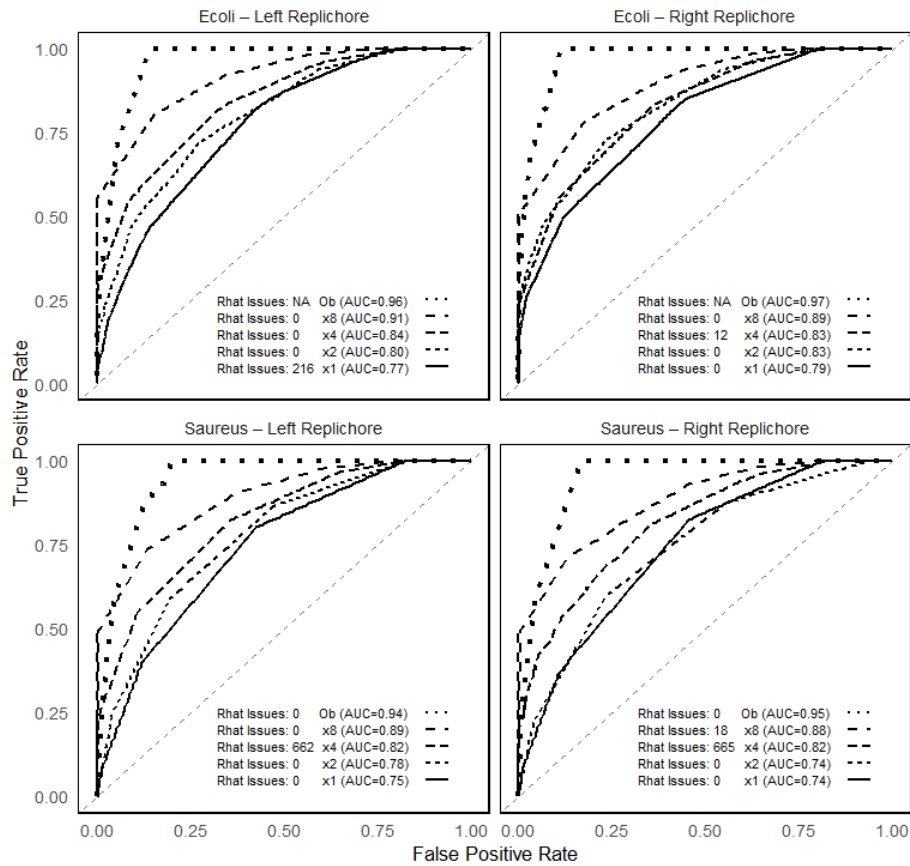


Figure 4.20. Plots of the ROC curves for the simulated datasets with varied cluster scale factors of 1, 2, 4 and 8. As the cluster size becomes longer, the AUC increases for all replichores.

impacting classification, in addition to the challenging shape of the data.

Table 4.7. Simulation Scenarios for Insertion Density Variation (Ess is essential and Non-Ess is non-essential, SD is standard deviation.)

Simulation	Mean (Ess)	SD (Ess)	Mean (Non-Ess)	SD (Non-Ess)	Description
1	1.0	1.0	1.0	1	Reference scenario
2	0.8	1.0	1.2	1	Reduced essential mean; increased non-essential mean
3	1.5	0.5	0.8	1	Increased essential mean with reduced SD; reduced non-essential mean

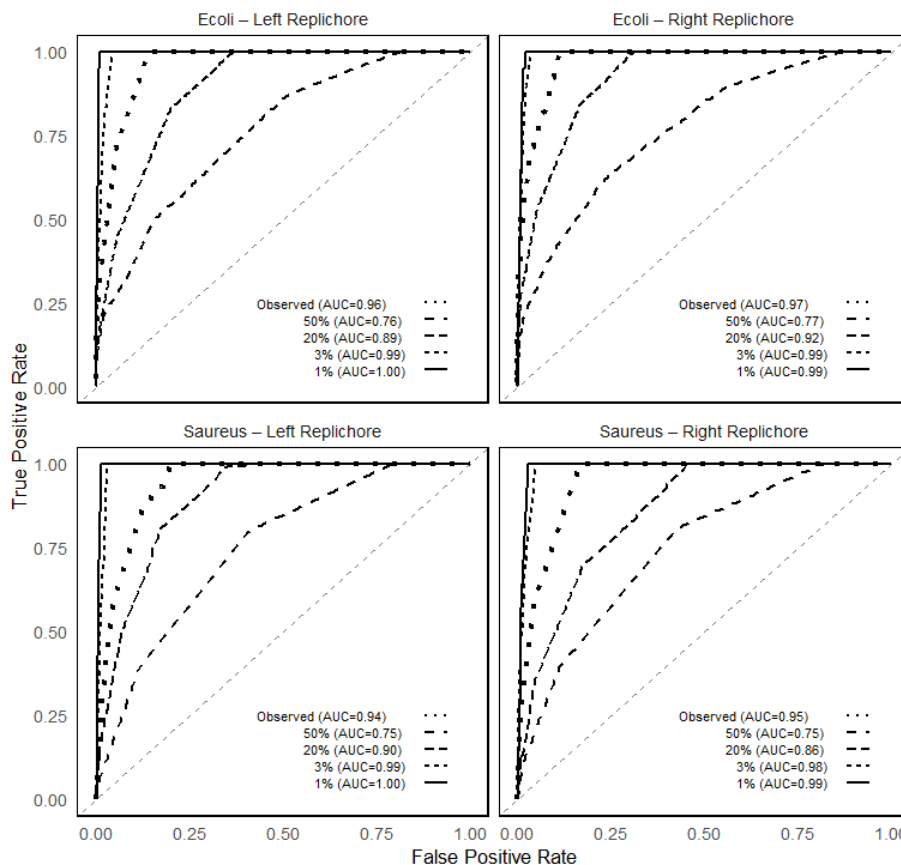


Figure 4.21. Plots of the ROC curves for the simulated datasets using different proportions of essential genes. As the proportion of essential genes drops, the classification performance improves.

4.7 Model Summary and Selection

Given the weaker results obtained when incorporating distance from the replication origin (DFRO) into the basic HMM, and the inconsistent patterns observed across genomes, the InsDens HMM is preferable for the classification of essential genes. The biological basis for insertion density being lower in essential genes is well established, providing a clear signal that can be effectively exploited. In contrast, the relationship between DFRO and essentiality is less robust, with variability between replichores, genomes, and environmental conditions. Combining both covariates was attempted, but required complex mixture distributions which provided highly unstable parameter estimate for no improvement in AUC.

The basic region HMM could also be used suitably, and offers a more stable alternative, with few or no convergence issues and broad applicability across datasets. One aspect that cannot be assessed thoroughly in this work is the generalizability, where the InsDens HMM may have some advantage with an

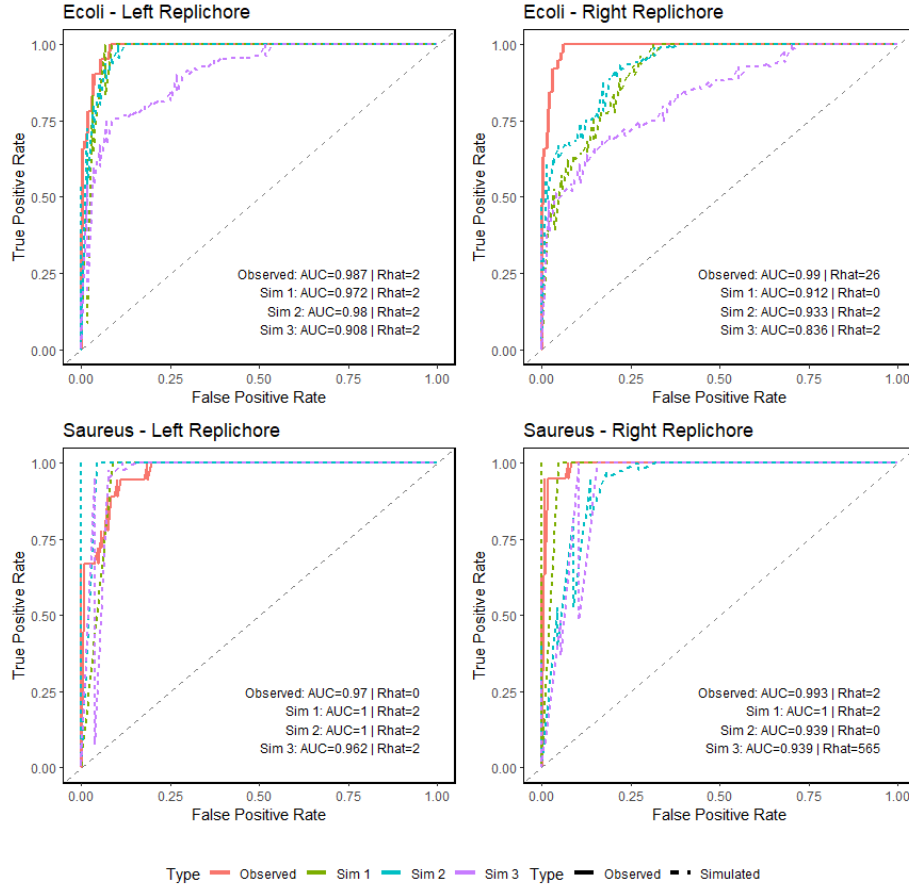


Figure 4.22. Plots of the ROC curves for the simulated (Sim) datasets with the insertion density distributions varied. The model used is the InsDens model. Performance reduces as the insertion density distributions between essential and non essential genes are moved closer together.

additional signal for classification. The downside of the InsDens model is an increase in instability, with some parameters failing to converge. This instability could be reduced through further tuning of priors and model adaptation.

Chapter 5

Conclusion

Characterizing the behavior of essential genes and developing accurate classifiers has significant potential in the advancement of biotechnology and the development of drug therapeutics. This dissertation has explored the two hypothesis that may contribute to essential gene classification: essential genes are more likely to form clusters, and that they tend to be located near the replication origin. These hypotheses were assessed using exploratory data analysis along with simulation and permutation testing, followed by the development of a HMM classifier for essential genes. This chapter summarizes the key outcomes of the investigation and evaluates the effectiveness of the proposed classification methods, while discussing their limitations and scope for improvement.

5.1 Dissertation Overview

Chapter 1 outlined the biological foundations of gene essentiality and its significance for therapeutic development. It distinguished between unconditionally essential genes, required for survival under all conditions, and conditionally essential genes, which depend on specific environments. The chapter also described experimental methods for identifying essential genes, including selective gene deletion and high-throughput mutagenesis approaches such as TraDIS, highlighting their advantages and limitations.

Chapter 2 detailed the *Escherichia Coli* (ecoli) and *Staphylococcus Aureus* (saureus) data sets used for the dissertation in addition to some exploratory data analysis evaluating the hypotheses. It highlighted the need to adapt the data produced sequentially to a segmented format that considers the genome organisation by replichore. Chapter 3 presented an overview of the methodologies applied to the data. A key method throughout was Monte Carlo testing, which was applied both in permutation tests, to assess whether observed cluster sizes and proximities to the replication origin could occur by random chance, and in the modeling phase using Bayesian Hidden Markov Models (HMMs). The chapter also explored prior application of HMMs in the

literature to sequential biological data where the aim is to classify essential genes. In Chapter 4 these approaches were applied to the *ecoli* and *saureus* datasets, split by replichore, and then explored the classification performance and robustness of the models through structured data simulation.

5.2 Main Results

The structure of the genomic data as a sequence of insertions introduces challenges for modelling and classifying essential genes. The raw data does not reflect the biological organization of bacterial chromosomes, which must be segmented first by genome and then split by replichore, since combining replichores after transformation would artificially duplicate genomic positions and distort clustering patterns. This retains the biological realism but exacerbates the class imbalance as essential genes are relatively rare. An additional complication of the data is the autocorrelation between the insertions, due to the clustering of essential genes. This violates independence assumptions of many statistical approaches which could lead to biased parameter estimates. Finally, the scope of these datasets is limited in both breadth and information density as the missing third genome restricts assessment of generalizability. The data is also limited to a small set of features, and ignores other important biological covariates such as GC content and operons (among others) which could provide additional signals for essentiality.

Clustering was assessed both by spatial proximity between essential genes and by run lengths of contiguous essential genes. Under both definitions, permutation testing provided strong evidence of clustering, with the observed data consistently exceeding the cluster size test statistics seen in a large number of simulated datasets. This finding supported the application of HMM's, which can exploit this for classification. On the other hand, the positional preference of essential genes along the genome was inconsistent. Only the right replichore of the *ecoli* dataset was located closer to the origin, whereas the *saureus* replichores were located further away and the left replichore of *ecoli* had no strong preference.

Due to memory limitations and parameter instability, the HMM was developed based on contiguous regions rather than individual genes. Each region is modeled as either essential or non-essential, with hidden states following a Markov chain, and the observed number of essential genes per region modeled using a Beta-Binomial (or Binomial with Beta prior) distribution. Weakly informative Dirichlet priors are applied to the transition probabilities to allow flexibility between states, and performance improves as region size decreases (AUC increases from 0.75 to 0.95 or higher). This approach reduces noise from uncertain gene-level models and allows practical computation time when incorporating additional covariates such as insertion density or distance from the origin. The mean posterior probabilities of the regions were used to develop a classifier for the true labels, with a flexible user defined threshold. While this model was reasonably

accurate, and did not exhibit any convergence issues for any parameters, there was still some variability in estimation and loss of resolution by using a gene level model. HMM's were preferred to other models due to their probabilistic output and wealth of biologically relevant information that is scarce in other classification tools. The model was also flexible in thresholding and biologically informed prior selection, in addition to having a low computational requirement.

The model was extended using a Hierarchical Bayes approach to include the insertion density and the distance from the origin. Given the varied behavior during the data analysis, and the poorer results when introducing it as a co-variate, the DFRO model was discarded in favor of the insertion density model. The insertion density behavior is generalizable across genomes (essential genes have lower densities), but the modeling found comparable or worse classification performance to the basic model with the additional cost of some instability in the parameters. Simulated data was generated from the features of the observed data, which is necessary to ensure the genomic features (such as clustering) are adequately represented. The basic HMM (without insertion density) demonstrated good performance in AUC and stability as features were varied, though the insertion density HMM had more erratic performance.

5.3 Limitations

The region based HMM was chosen as a compromise on computational efficiency, resolution and biologically interpretable outputs. The initial phase of developing a classifier included multiple classification approaches, including Gaussian Process, K-Nearest Neighbors, Random Forests and Logistic Regression. These were not included in the report, but in some cases these models provided higher classification performance than the HMM but with less interpretability of the outputs. Viewed purely as a classification exercise, these models would have sufficed with the inclusion of the insertion density and distance from the replication origin (code for the attempts are included in Chapter 6). Gene level models were also explored but are computationally intensive and practically unfeasible on used hardware when introducing the additional covariates. Refining the HMM to include gene level granularity would have more accurate classifications but require additional computational resources. Since the distance from the origin is not a strongly distinct feature, like with insertion density, convergence or parameters was challenging throughout and large region sizes tended towards instability. Finally, the HMM's developed are first-order so it only accounts for dependencies between immediate neighbors and longer-range patterns such as operons are only captured indirectly through high self-transition probabilities.

5.4 Recommendations for Further Work

Future research could refine the current region-based HMM by developing gene-level models, which may provide finer resolution and predictive accuracy offset

by computational cost. Additional intermediate states (Chao, M, et al, 2013) would provide more granularity to gene regions that are neither wholly essential or non-essential, which would reduce false positives and add nuance to the classification. This approach could also facilitate the identification of conditionally essential genes that are context-dependent or sensitive to the environment. Predictive accuracy of the HMM could be improved by combining deep learning approaches, such as convolutional and recurrent neural networks, with the probabilistic framework of the HMM similar to the Tiberius model (Gabriel, L, et al, 2024). Integrating additional genomic covariates such as gene GC content or operon structure would reveal the importance of features within the models and determine those that can be generalized. Other modelling approaches to HMMs could also be examined, such as Conditional Random Fields (CRFs), which have the advantage of incorporating multiple genomic features simultaneously, handling dependencies between sites, and providing probabilistic outputs for each gene or region, potentially improving classification accuracy and interpretability over HMMs (DeCaprio, D, et al, 2007).

Chapter 6

Appendix

6.1 Declaration of Ethics Approval

The research ethics approval process was completed in August 2024 under the supervision of the original project advisor, prior to the commencement of any dissertation work. As the study does not involve any personal or identifiable data, formal ethics approval was not required beyond supervisor sign-off (shown in Figure 6.1).

Form 1: Student declaration (for research that does not involve human participation or analysis of secondary data)

School of Mathematics and Statistics
Research Ethics Review for Postgraduate Taught Students

Dissertation title: Investigating Properties of Bacterial Genes

In signing this student declaration, I am confirming that:

My project does **not** involve people participating in research either directly (e.g. interviews, questionnaires) and/or indirectly (e.g. people permitting access to data).

My project does not therefore require an ethics review, and I have not submitted a Research Ethics Application Form.

Student registration number: 230147587 Date: 1st August 2024

Name of supervisor: *Dr. Kevin Walters* Date: 1st August 2024

Figure 6.1. Ethics Approval document required for initiation of the dissertation.

6.2 Note to the Examiner

The supervisor who developed the project scope left in February 2025, after which supervisory support continued under a new advisor. While technical

guidance in statistical genetics was no longer available beyond the Basic Region HMM, ongoing advice was provided on mathematical aspects following the modeling and simulation stage. Alternative classification approaches outside of Hidden Markov Models were considered, though their use was discouraged for reasons outlined in the preceding chapters. Excluding the Appendix content, the main matter of the report is 65 pages (with the approval excerpt) and an additional 5 pages of references.

6.3 Functions List

There are many functions developed with most being too long to include in the main text or Appendices without overwhelming the space of the report (which is already heavily truncated). Pseudo code would limit the scope of the main text, so descriptions are provided where appropriate. The code comprises the following: data preprocessing, model fitting, result visualization and data simulation. Key functions are provided as references to a Github repository (suggested by the supervisor). All code is developed independently and only functions that are novel or integral for key parts of the dissertation are included.

6.4 ACF Plots of Basic Regions HMM

6.5 Cross Validation

Table 6.1. Summary of Dissertation functions with Descriptions and Reference links.

Section	Function	Description	GitHub Link
Chapter 2	Data Cleaning	Preprocessing raw datasets	Link
Chapter 2	Distance from Replication Origin	Computes gene and region distances from origin	Link
Chapter 4	Mean Distance and Permutation p Value	Mean distance and permutation test	Link
Chapter 4	Median Distance and Permutation p Value	Median distance and permutation test	Link
Chapter 4	Simulated Mean Cluster Size and p Value	Simulated Mean cluster size test	Link
Chapter 4	Simulated Max Cluster Size and p Value	Simulated max cluster size test	Link
Chapter 4	Gene Level HMM Stan Model	Bayesian Gene level HMM	Link
Chapter 4	Region HMM Basic	Baseline HMM for regions	Link
Chapter 4	Region HMM with Insertion Density	HMM incorporating insertion density	Link
Chapter 4	Region HMM with Distance from Origin	HMM incorporating distance from origin	Link
Chapter 4	Threshold and Cost Functions	Utility functions for thresholds and costs	Link
Chapter 4	Data Simulation Functions	Simulates datasets for model testing	Link
Appendix	Cross Validation for Indens Model	CV for insertion density model	Link
Appendix	kNN Classifier	k nearest neighbors classifier	Link
Appendix	Random Forest Classifier	Random forest classifier	Link
Appendix	Gaussian Process Classifier	Gaussian process classifier	Link

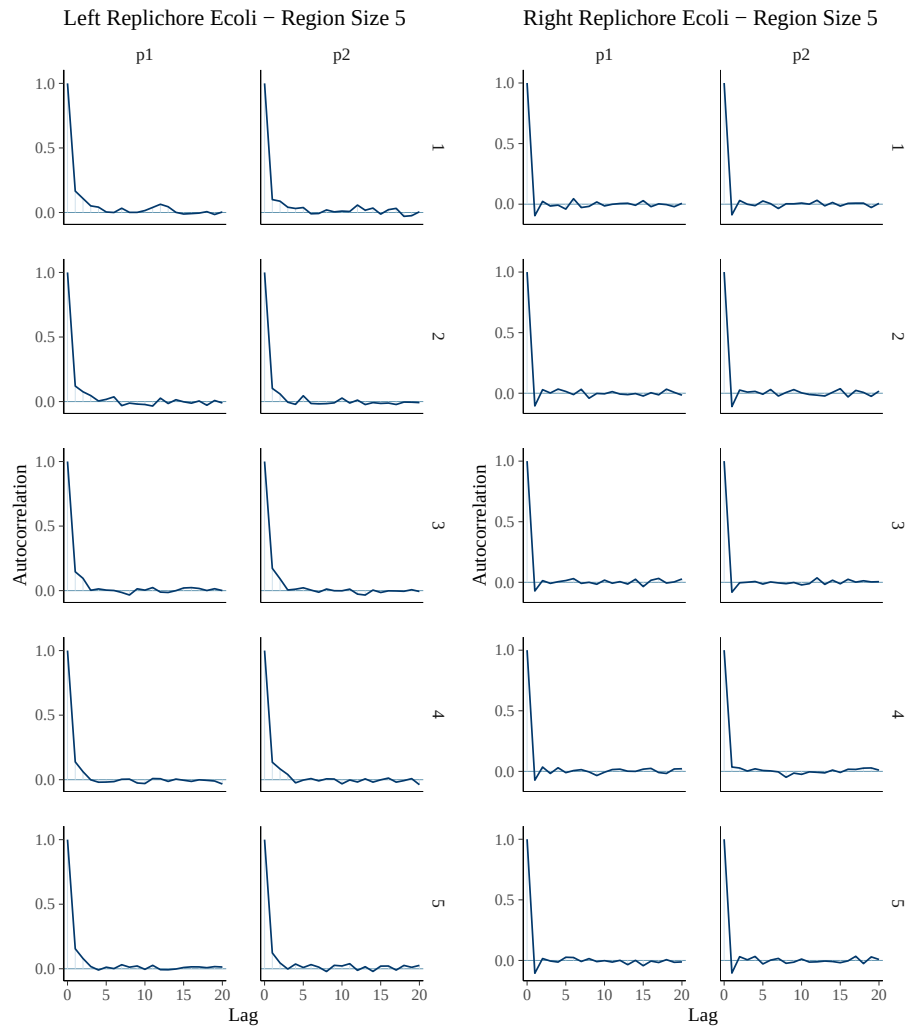


Figure 6.2. ACF Plots for the left replicore of Ecoli, for $p1$ and $p2$.

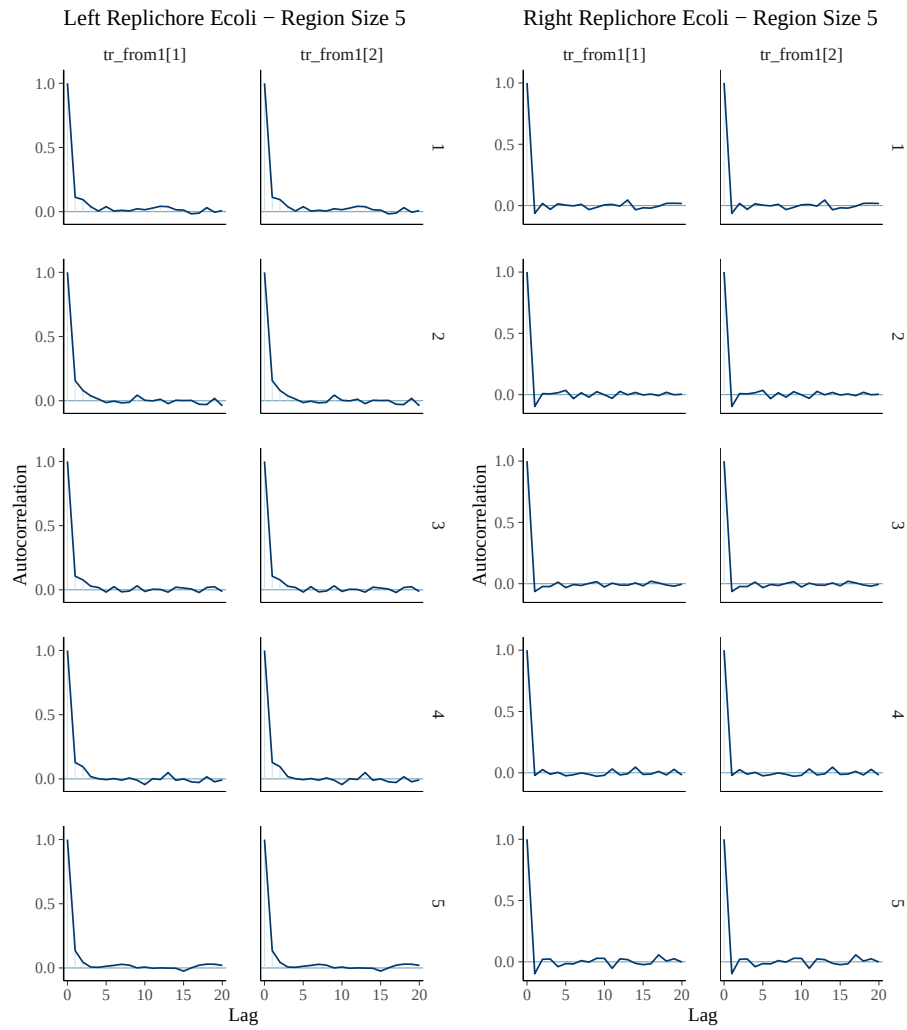


Figure 6.3. ACF Plots for the left replicore of Ecoli, for trfrom1[1] and trfrom1[2].

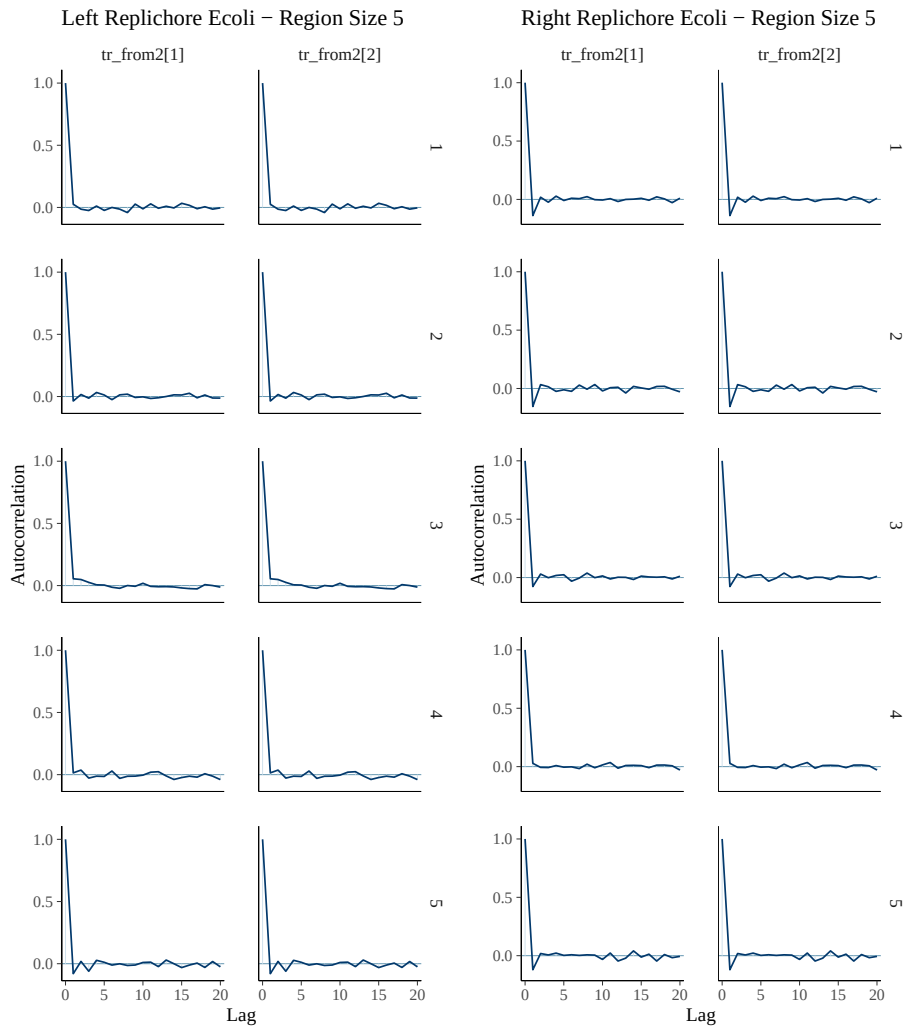


Figure 6.4. ACF Plots for the left replicore of Ecoli, for $\text{trfrom2}[1]$ and $\text{trfrom2}[2]$.

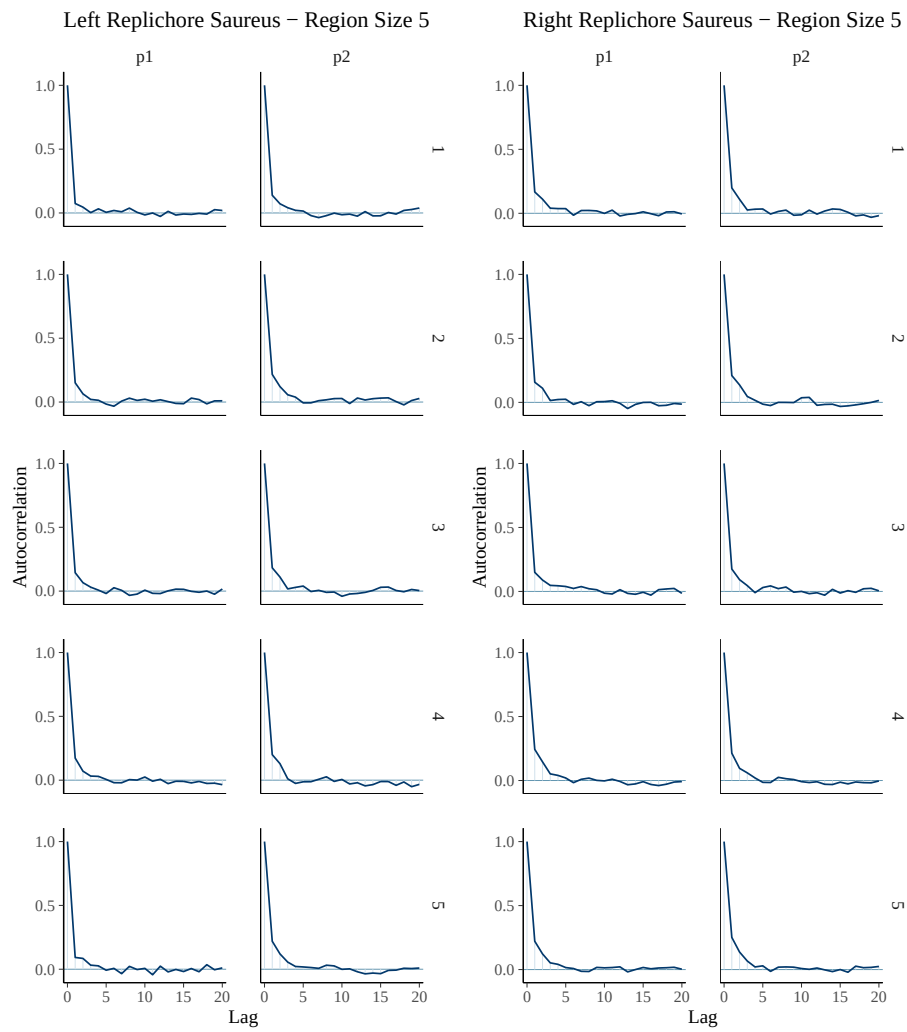


Figure 6.5. ACF Plots for the left replicore of *Saureus*, for $p1$ and $p2$.

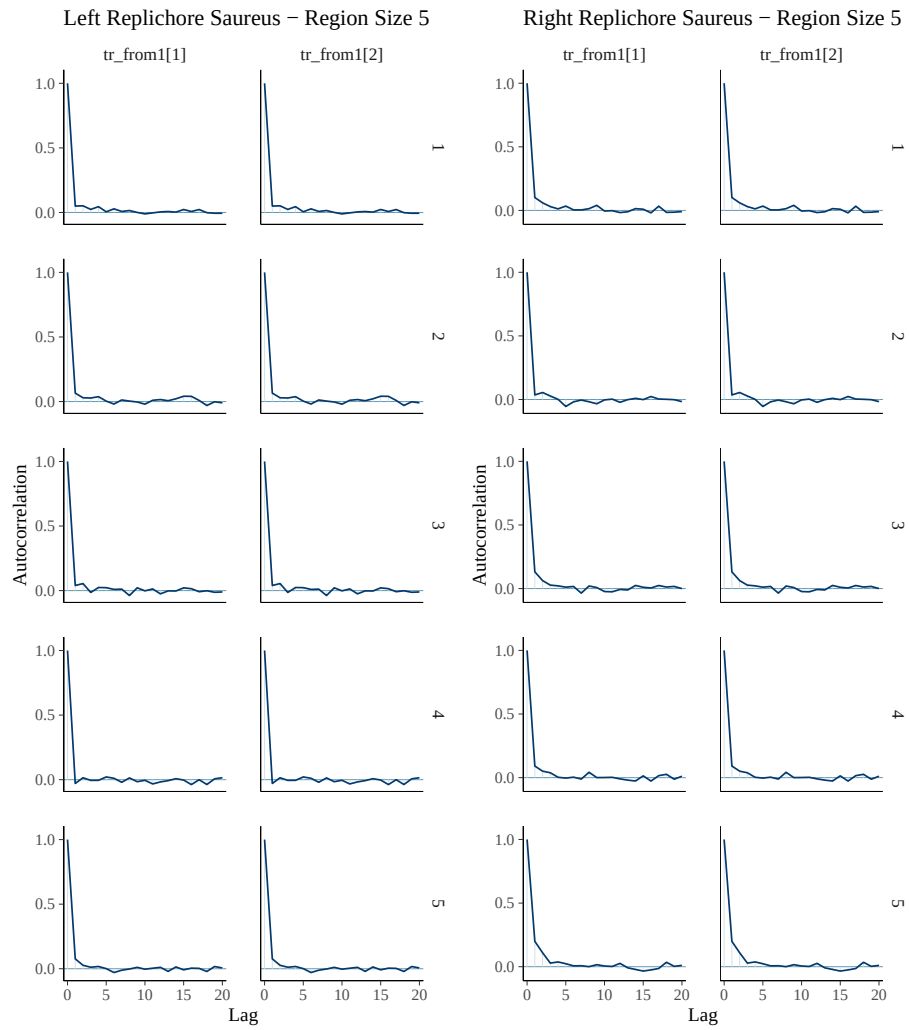


Figure 6.6. ACF Plots for the left replichore of Saureus, for `trfrom1[1]` and `trfrom1[2]`.

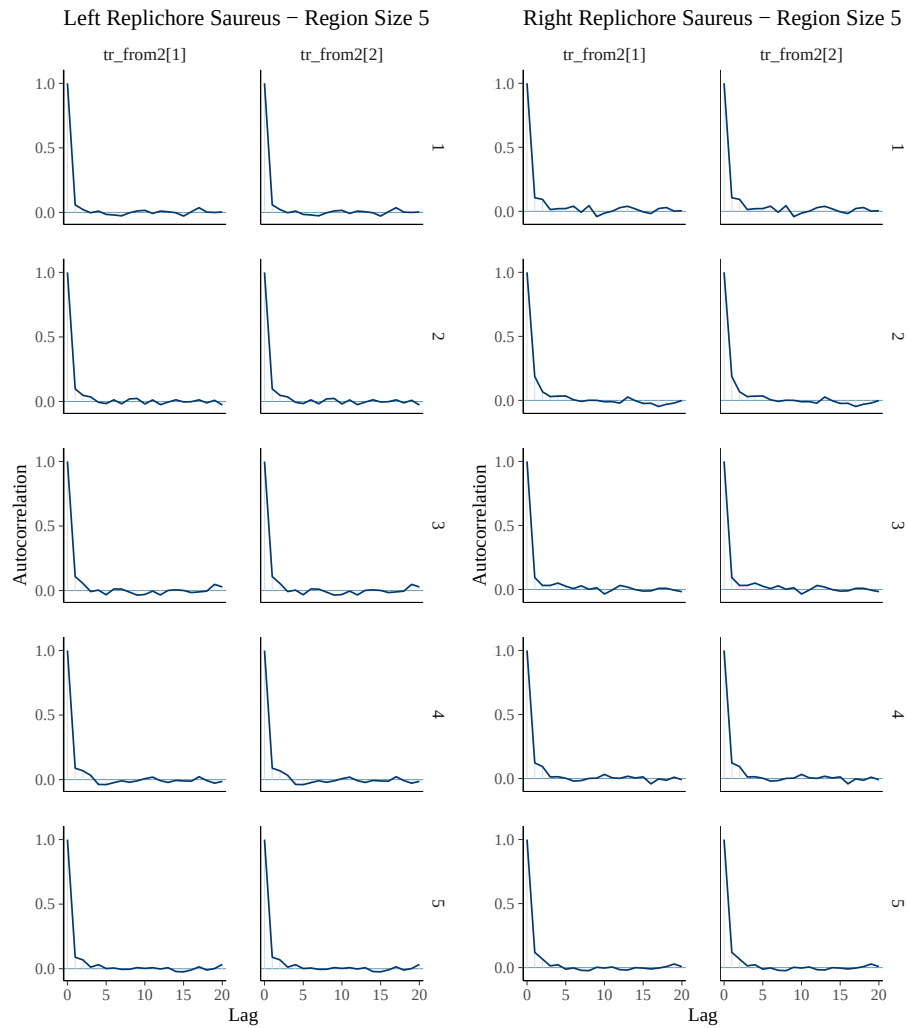


Figure 6.7. ACF Plots for the left replicore of Saureus, for trfrom2[1] and trfrom2[2].

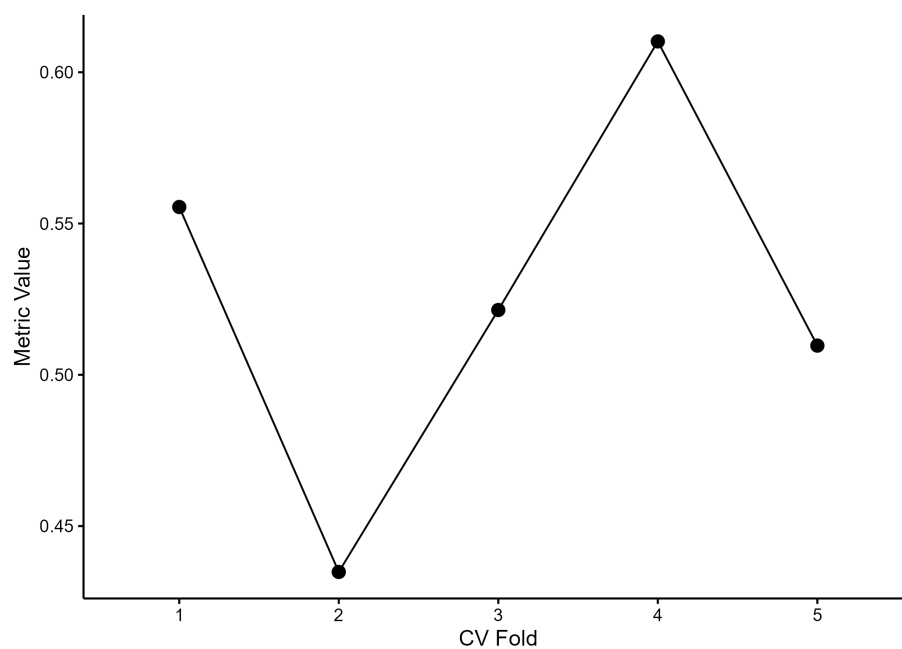


Figure 6.8. Plot of the AUC as a function of the cross validation folds for the left replicore of the *ecoli* genome. See Github repository for the function.

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